CASE STUDY

The Year 2010 in Intrusions: A Dated Record

A TeamStarWolf case study. Published 13 November 2026; every source checked as of that date.

Executive summary

The year opened with a disclosure about the year before. On 12 January Google said that in mid-December 2009 it had detected a highly sophisticated, targeted attack on its corporate infrastructure originating from China that resulted in the theft of intellectual property, that at least twenty other large companies in the internet, finance, technology, media and chemical sectors had been similarly targeted, and that a primary goal appeared to be the Gmail accounts of Chinese human rights activists, only two of which appear to have been accessed (Confirmed [1]). Microsoft's Advisory 979352 followed on 14 January, NVD published CVE-2010-0249 on 15 January as an Internet Explorer use-after-free "exploited in the wild in December 2009 and January 2010 during Operation Aurora", and MS10-002 fixed it on 21 January (Confirmed [3][5][4]). The same day the Secretary of State said that countries or individuals that engage in cyber attacks should face consequences and international condemnation (Confirmed [9]). On 22 March Google stopped censoring search on Google.cn and redirected users to Google.com.hk (Confirmed [2]). ATT&CK says the Elderwood group was "reportedly responsible" (Confirmed that ATT&CK says so [11]; the attribution is Reported). CISA added CVE-2010-0249 to its Known Exploited Vulnerabilities catalog on 20 May 2026 (Confirmed [8]). The Zero-Day Trade study is the home for Aurora.

The year's defining malware arrived on a USB stick. Symantec's dossier records that VirusBlokAda reported the malware on 17 June; Microsoft's Advisory 2286198 on the Windows shortcut flaw CVE-2010-2568 followed on 16 July, ICS-CERT said the trojan interacted with Siemens SIMATIC WinCC and STEP 7, and out-of-band bulletin MS10-046 shipped on 2 August, when "a number of malware families" were already exploiting the flaw (Confirmed [30][24][22][25]). By the time Symantec published its analysis on 30 September it counted about 100,000 infected hosts, roughly 60 percent of them in Iran (Confirmed [30]). The Stuxnet study is the home; this chronicle records the 2010 dates and the defender record.

Crime was prosecuted at scale and in public. Max Butler was sentenced to 13 years on 12 February, Albert Gonzalez to 20 years on 25 March, and a federal judge granted Microsoft a restraining order on 22 February that cut off 277 domains of the Waledac spam botnet (Confirmed [13][66][15]). Spanish and Slovenian police arrested the suspected Mariposa operators and creator (Confirmed [16]). On 30 September prosecutors in New York charged 37 defendants over Zeus-enabled bank fraud, and the next day the FBI said a Zeus-based ring had stolen USD 70 million and attempted about USD 220 million (Confirmed [42][43][44]). In December, as later charged, Anonymous ran denial-of-service attacks on PayPal after it suspended the WikiLeaks account (Confirmed as charged [46]).

Espionage was documented by researchers and a congressional commission, and neither made a finding of state responsibility; the commission offered an assessment, not a finding (Confirmed [18][10]). "Shadows in the Cloud" of 6 April documented a network that compromised systems in India's national security establishment, the Offices of the Dalai Lama and the United Nations, and placed the attackers in Chengdu only by circumstantial evidence (Confirmed [18]; the location is Reported). The US-China Economic and Security Review Commission said China Telecom advertised erroneous routes for about 18 minutes on 8 April that sent traffic for about 15 percent of internet destinations through Chinese servers, and that it could not determine what, if anything, was done with the traffic (Confirmed [10]). On 1 September the Deputy Secretary of Defense disclosed that an infected flash drive had compromised US Central Command networks in 2008 (Confirmed [38]).

The web's defaults moved. Gmail began turning on HTTPS by default for all users on 12 January, Chromium launched a vulnerability reward program on 28 January, RFC 5746 fixed TLS renegotiation in February, HTTPS Everywhere shipped on 17 June, the signed DNS root's first trust anchor became valid on 15 July, and on 24 October Firesheep showed how easily session cookies could be taken on open Wi-Fi (Confirmed [76][74][70][77][73][68]). Governments organized: NIST published SP 800-37 Revision 1 in February, US Cyber Command began operations in May, NIST published its smart grid guidelines on 31 August, DHS ran Cyber Storm III from 27 September, and the UK committed GBP 650 million of new funds to cyber security on 19 October (Confirmed [69][38][78][79][80]).

This study records the year by date. It is built from sources opened on 12 November 2026 (except source 82, which is carried from the E-Commerce Threats study) and from checks of the library's data run on 12 and 13 November 2026, and is checked as of 13 November 2026.

Key facts at a glance

Field	Value
Period	1 January to 31 December 2010, plus the records that open, close or explain a 2010 event: Google's December 2009 detection, the 2008 CENTCOM flash-drive compromise disclosed in 2010, the 2011 charging releases for the AT&T iPad exposure and the PayPal attacks, Symantec's dossier v1.4 (February 2011), and KEV additions from 2021 to 2026
The headline disclosure	Aurora: Google's post (12 Jan); Advisory 979352 (14 Jan); NVD CVE-2010-0249 (15 Jan); MS10-002 and the Clinton speech (21 Jan); Google.cn redirect (22 Mar) (Confirmed [1][3][5][4][9][2])
Stuxnet in 2010	VirusBlokAda report (17 Jun); CERT/CC VU#940193 (15 Jul); Advisory 2286198 and US-CERT alert (16 Jul); MS10-046 (2 Aug); MS10-061 (14 Sep); Symantec dossier (30 Sep) (Confirmed [30][26][24][23][25][31])
Exploited at or before the fix	Microsoft CVE-2010-0249, -0806, -1885, -2568, -2729, -2743, -3332, -3338, -3962; Mozilla CVE-2010-3765; Adobe CVE-2010-1297, -2883, -3654 (Confirmed [4][52][21][25][31][53][39][55][45][56][60][62][63])
Microsoft's year	MS10-001 (12 Jan) to MS10-106 (14 Dec): 106 bulletins by numbering; October alone carried 16 bulletins with 49 vulnerability sections (Confirmed [49][50]; counts Inferred)
KEV tail	23 CVE-2010 entries in catalog 2026.11.10, four flagged for known ransomware use; CVE-2010-0249 and CVE-2010-0806 added on 20 May 2026 (Confirmed [8])
Crime and prosecutions	Butler (12 Feb); Waledac order (22 Feb); Gonzalez (25 Mar) and Scott (29 Mar); Mariposa arrests (Feb and Jul); Zeus charges (30 Sep) and Trident Breach (1 Oct); PayPal attacks (6 to 10 Dec, charged 2011) (Confirmed [13][15][66][67][16][42][44][46])
State and espionage	Aurora (detected Dec 2009); Shadows in the Cloud (6 Apr); China Telecom routes (8 Apr); Night Dragon (Nov 2009 to Feb 2011); Buckshot Yankee disclosed (1 Sep) (Confirmed [1][18][10][47][38])
Web and open source	Apache JIRA attack (5 to 9 Apr); AT&T iPad enumeration (5 to 9 Jun); DLL preloading advisory (23 Aug); ASP.NET padding oracle (17 Sep); Twitter onMouseOver (21 Sep); Firesheep (24 Oct) (Confirmed [17][20][37][39][41][68])
Defaults and incentives	Gmail HTTPS by default (12 Jan); Chromium rewards (28 Jan); RFC 5746 (Feb); HTTPS Everywhere (17 Jun); KSK-2010 valid (15 Jul); Mozilla USD 3,000 bounty (15 Jul) (Confirmed [76][74][70][77][73][75])
Law, policy and standards	SP 800-37 Rev. 1 (Feb); CWE/SANS Top 25 (29 Mar); OWASP Top 10 (Apr); Cyber Command operating (May); NISTIR 7628 (31 Aug); Cyber Storm III (27 Sep to 1 Oct); UK SDSR (19 Oct) (Confirmed [69][71][72][38][78][79][80])
Written later	ATT&CK's Flame (S0143) "active since at least 2010", disclosed in 2012; Operation Dust Storm (C0016) first seen January 2010 (Confirmed [48]; Demonstrated [84])
This study's MITRE binding	71 ATT&CK Enterprise techniques in 88 technique-tactic entries: 58 listed by ATT&CK for Elderwood (G0066), Hydraq (S0203), the Night Dragon campaign (C0002) or Stuxnet (S0603), seven described by a primary and assigned here, two from Reported sources, four inferred; ten ATT&CK for ICS techniques listed for Stuxnet; CWE ids from NVD and KEV; CAPEC and D3FEND pairings from the library's data; an Enterprise and an ATT&CK for ICS Navigator layer (Part 9)

Key findings

- **Aurora is a 2010 record of a 2009 attack.** Google detected it in mid-December 2009 and disclosed it on 12 January 2010; the advisory, the CVE record, the fix and the policy response all date from January 2010 (Confirmed [1][3][5][4][9]).
- **The fix came a week after the advisory, and KEV came sixteen years later.** Advisory 979352 is dated 14 January and MS10-002 21 January; KEV added CVE-2010-0249 on 20 May 2026 (Confirmed [3][4][8]).
- **Removable media reached the control room.** The shortcut flaw ran code when a USB drive was viewed in a file manager, without AutoRun, and the first malware to use it went after Siemens SIMATIC software (Confirmed [22][23]).
- **There was no fix for at least 17 days.** Advisory 2286198 came on 16 July, six days after CERT/CC's date public, and MS10-046 on 2 August; the workarounds MS10-046 records, and ICS-CERT's advice, were to turn off shortcut icon display and the WebClient service and to block shortcut files and outbound SMB (Confirmed [24][26][25][22]; the day count is Inferred).
- **Documents and browsers were exploited all year.** Internet Explorer, Firefox, Reader and Flash flaws were exploited at or before their fixes from January to December (Confirmed [4][52][45][56][60][62][63]).
- **Prosecution caught up with the carding era.** Butler was sentenced to 13 years in February and Gonzalez to 20 years in March, and New York charged 37 Zeus defendants in a single day in September (Confirmed [13][66][42]).
- **Takedowns moved into civil court.** Microsoft cut Waledac's domains through a temporary restraining order against John Does, not a criminal case (Confirmed [15]).
- **Researchers and a commission documented the espionage without a state finding.** Shadows in the Cloud names no state, and the USCC offers an assessment of Chinese government knowledge, not a finding (Confirmed [18][10]).
- **Encryption and integrity defaults moved in 2010.** Gmail's default HTTPS, HTTPS Everywhere, RFC 5746 and the signed root all date from this year, and Firesheep showed what was left unencrypted (Confirmed [76][77][70][73][68]).
- **KEV is still writing 2010 down.** Two 2010 ids entered KEV in 2026 and two in 2025; four of the 23 are flagged for known ransomware use (Confirmed [8]).

Top recommendations

1. **Treat removable media as an entry point, not a convenience.** The 2010 shortcut flaw needed only a drive viewed in a file manager, and ICS-CERT's advice was to disable AutoRun, restrict privileges and be cautious with USB drives (Confirmed [22][23]).
2. **Apply the workaround while the fix is pending.** For at least 17 days after Advisory 2286198 the shortcut flaw had no fix, and the defenses MS10-046 and ICS-CERT list are workarounds: disable shortcut icons, WebClient and AutoRun (Confirmed [24][25][22]; Inferred for the count).
3. **Treat an advisory that says "limited, targeted attacks" as an emergency.** Advisories 979352, 2286198, 2416728 and 2458511 all preceded fixes for flaws already in use (Confirmed [3][24][39][45]).
4. **Encrypt sessions end to end, not just the login.** Firesheep took cookies that were sent in clear after an encrypted login, and its author argued that only full HTTPS fixes it (Confirmed [68]).

5. **Change default and hard-coded credentials in control software.** NVD says Stuxnet used a hard-coded password in Siemens WinCC and PCS 7 (Confirmed [33]).
6. **Patch the public-facing middleware you forgot.** JBoss, ColdFusion and Exim flaws from 2010 are in KEV, and the JBoss and ColdFusion entries are flagged for known ransomware use (Confirmed [8][59][61][64]).
7. **Watch for credential capture at online banking login.** Zeus recorded keystrokes at login and money mules moved the funds (Confirmed [42][43]).
8. **Lock down issue trackers and admin consoles.** Apache's JIRA attack combined a cross-site scripting link with a brute-force login on the same host (Confirmed [17]).
9. **Do not read KEV dates as event dates.** CVE-2010-0249 was exploited in 2009 and 2010 and listed in 2026 (Confirmed [5][8]).
10. **Read "originating from China" for what it says.** Google, the USCC and the Shadows report each stop short of a state finding (Confirmed [1][10][18]).

Scope and method

This study covers intrusions, vulnerabilities, malware, botnets, card theft and fraud, prosecutions, state and espionage operations, defaults and incentives, regulation, standards and policy dated to 2010, plus the later records that open, close or explain a 2010 event. It follows the method of the case studies collection: primary sources first, every claim graded, and numbers given as the source states them. Every external source except source 82, which is carried from the E-Commerce Threats study, was opened on 12 November 2026, and the library data checks were run on 12 and 13 November 2026. The Intelligence Hub feed, which carries 23 KEV rows for CVE-2010 ids and a historic row for the 16 July shortcut advisory, was used as a map to find primaries, not as a citation for any event (Confirmed [83]).

Grade	Meaning in this study
Confirmed	The primary record was opened and states the fact: a vendor bulletin or advisory, a CERT/CC note, a US-CERT, ICS-CERT or CISA record, a CVE or NVD record, the KEV catalog, an FBI or US Attorney release, a court order as the plaintiff describes it, a congressional commission report, a government strategy or exercise report, an official's own statement, a researchers' or vendor's report stated as its own finding, a NIST, IETF, IANA, MITRE or OWASP record, a live ATT&CK page, or a library study, only for the fact that it is the home for an event
Reported	Only a secondary source supports the fact, or a source relays someone else's claim: World Nuclear News relaying Reuters, the Christian Science Monitor, the USCC quoting a vendor or Google's security manager, an FBI release quoting an industry estimate, and ATT&CK's "reportedly" attributions. The source is named in the text
Demonstrated	A check this study ran itself and saw the result (counts in the KEV catalog; id, name, tactic and page checks in the library's data files and `mitre/` page tree), or a public proof whose author shows the effect (Firesheep)
Inferred	The study's own reading, labelled where it appears. Every ATT&CK, CAPEC and D3FEND id bound to a 2010 event is Inferred as to the binding unless ATT&CK lists it for the object, and so is every CWE that NVD or KEV does not list

Charges, complaints and indictments are allegations; where this study cites one, the record confirms that the charge was made, not that it was proven. Company statements are the company's own account, and vendor and researcher attributions are those authors' assessments, not government findings.

Limits a reader should know:

- **CVE existed; ATT&CK and KEV did not.** Every 2010 product flaw here carries a CVE id. No ATT&CK mapping existed in 2010, so every technique id in Part 9 is assigned in hindsight from later ATT&CK records or by this study (Inferred [5][11]). KEV's 2010 entries were added from November 2021 to May 2026 (Confirmed [8]).
- **Some events have no id of any kind.** The China Telecom route event, the Waledac takedown and the Clinton speech carry no CVE and no ATT&CK object, and the USCC makes no finding of intent for the routes (Confirmed [10][15][9]).
- **ICS-CERT's first Stuxnet advisory carries no original date on the page opened.** ICSA-10-201-01C shows only its 2014 revision, so this study dates its events from VirusBlokAda's report, CERT/CC and Microsoft (Confirmed [22][26][24]).
- **What was not checked.** No primary was opened, or the page would not open, for: the Justice Department's own release on the Gonzalez sentencing (the US Attorney's release of 16 April was used instead); Adobe's 2010 advisories and bulletins (APSA10-01, APSA10-02, APSA10-05, APSB10-07, APSB10-14 and APSB10-21) and its January 2010 post about its corporate network; the FTC's 2010 orders in the Twitter and Rite Aid matters; Massachusetts 201 CMR 17.00; the US Cyber Command history page and its exact initial and full operating dates; the DHS and DoD cybersecurity memorandum of October 2010; the Dutch Bredolab takedown; the UK Information Commissioner's first monetary penalties; the June 2010 arrests of ten alleged Russian agents; the "Here you have" mass-mailer; the Gawker breach; Koobface; PCI DSS 2.0; ISO/IEC 27003:2010; EMET 2.0; Symantec's Elderwood report and McAfee's Aurora analysis; the declassification date of the CNCI summary; Microsoft's October 2010 bulletin summary page; and a total count of 2010 CVE records. They are left out or marked "not checked" where they touch a verified fact. Their absence here is not evidence that nothing happened.

Part 1: the year as a dated timeline

One row per event or report. The date is the event date where the source gives one, otherwise the publication date. Rows marked "frame" or "boundary" open or close a 2010 event.

Date	Event	Thread	Sector	Grade	Sources
2008 (frame)	An infected flash drive compromises CENTCOM classified and unclassified networks; Operation Buckshot Yankee responds (disclosed 1 Sep 2010)	State	Defense	Confirmed (official's statement)	[38]
2009-11 to 2011-02 (frame)	ATT&CK's Night Dragon campaign against oil, energy and petrochemical companies. Home: China-attributed threat actors	State (China-based, per researchers)	Energy	Confirmed; attribution Reported	[47]

Date	Event	Thread	Sector	Grade	Sources
2009-12 (frame)	Google detects a targeted attack on its corporate infrastructure (Operation Aurora); CVE-2010-0249 exploited in the wild from December	State (reportedly Elderwood)	Technology and others	Confirmed; attribution Reported	[1][5][11]
2010-01	ATT&CK's Operation Dust Storm (C0016) first seen	Espionage (unattributed)	Multiple	Demonstrated	[84]
2010-01-08	Heartland and Visa announce a settlement of up to USD 60 million for Visa issuers, as carried by the home study. Home: E-Commerce Threats	Card theft	Payments	Inferred (carried)	[82][87]
2010-01-12	Google discloses the attack; Gmail turns HTTPS on by default; MS10-001, the year's first bulletin	State; Defaults; Vulnerability	Technology	Confirmed	[1][76][49]
2010-01-14	Microsoft Advisory 979352 and CERT/CC VU#492515 on the Internet Explorer flaw; version 1.1 on 15 Jan cites limited targeted attacks	Vulnerability	Cross-sector (browsers)	Confirmed	[3][7]
2010-01-15	NVD and the CVE Program publish CVE-2010-0249, "exploited in the wild ... during Operation Aurora"	Vulnerability	Cross-sector	Confirmed	[5][6]
2010-01-21	MS10-002 fixes CVE-2010-0249 and seven other flaws; Secretary Clinton's Remarks on Internet Freedom. Home: Zero-Day Trade	Vulnerability; Policy	Cross-sector; government	Confirmed	[4][9]
2010-01-28	Chromium announces its vulnerability reward program	Incentives	Technology	Confirmed	[74]
2010-02	NIST SP 800-37 Revision 1; RFC 5746, TLS Renegotiation Indication Extension	Standards	Government; cross-sector	Confirmed	[69][70]
2010-02	Spanish Guardia Civil arrests three suspected Mariposa operators	Crime	Consumers, finance	Confirmed	[16]
2010-02-09	MS10-015 fixes the publicly disclosed kernel flaw CVE-2010-0232, with no exploitation known at release	Vulnerability	Cross-sector	Confirmed	[51]
2010-02-12	Max Ray Butler sentenced to 13 years and USD 27.5 million restitution. Home: Four Books on Cybercrime	Crime	Finance (carding)	Confirmed	[13]
2010-02-21	CVE-2010-0188 (Adobe Reader and Acrobat before 9.3.1) published	Vulnerability	Cross-sector	Confirmed	[58]

Date	Event	Thread	Sector	Grade	Sources
2010-02-22	E.D. Va. grants Microsoft's restraining order against the Waledac botnet's 277 domains (announced 24 Feb)	Botnet takedown	Technology, consumers	Confirmed (plaintiff's account)	[15]
2010-03-25	Albert Gonzalez sentenced to 20 years. Home: E-Commerce Threats	Crime	Retail, payments	Confirmed	[66]
2010-03-29	2010 CWE/SANS Top 25; TJX conspirator Christopher Scott sentenced to seven years	Standards; Crime	Cross-sector; retail	Confirmed	[71][67]
2010-03-30	MS10-018 fixes CVE-2010-0806, which attackers were attempting to exploit	Vulnerability	Cross-sector (browsers)	Confirmed	[52]
2010-04	OWASP Top 10 2010 (release day from the PDF's metadata, approximate)	Standards	Web	Confirmed	[72]
2010-04-05 to 2010-04-09	Targeted attack on the Apache Software Foundation's JIRA host (reported 13 Apr)	Intrusion	Open-source software	Confirmed	[17]
2010-04-06	"Shadows in the Cloud" published (Citizen Lab op-ed on 7 Apr)	State (unattributed)	Government, NGO	Confirmed	[18][19]
2010-04-08	China Telecom advertises erroneous routes for about 18 minutes	Routing	Government, telecom	Confirmed (USCC's account)	[10]
2010-04-28	CVE-2010-0738, JBoss JMX-Console access control, published	Vulnerability	Cross-sector (servers)	Confirmed	[59]
2010-05	US Cyber Command begins operations under STRATCOM, per the Deputy Secretary of Defense	Policy	Defense	Confirmed (official's statement)	[38]
2010-06-05 to 2010-06-09	A script mimicking an iPad 3G harvests about 120,000 ICC-ID and e-mail pairs from AT&T's servers (charged 18 Jan 2011)	Exposure	Telecom	Confirmed (as charged)	[20]
2010-06-08	CVE-2010-1297 (Flash, Reader, Acrobat) published; exploited in the wild in June	Vulnerability	Cross-sector	Confirmed	[60]
2010-06-17	VirusBlokAda reports the malware later named Stuxnet; EFF and the Tor Project release HTTPS Everywhere. Home: Stuxnet	State; Defaults	Industrial; web	Confirmed	[30][77]
2010-07-10	Date the shortcut flaw became public, per CERT/CC	Vulnerability	Cross-sector	Confirmed	[26]
2010-07-13	MS10-042 fixes CVE-2010-1885 under active attack	Vulnerability	Cross-sector	Confirmed	[21]

Date	Event	Thread	Sector	Grade	Sources
2010-07-15	CERT/CC VU#940193; KSK-2010 root trust anchor valid from this date; Mozilla raises its bounty to USD 3,000	Vulnerability; Defaults; Incentives	Cross-sector; internet	Confirmed	[26][73][75]
2010-07-16	Microsoft Advisory 2286198 and US-CERT alert on the shortcut flaw; Realtek certificate revoked, per Symantec	Vulnerability	Cross-sector	Confirmed	[24][23][30]
2010-07-19	Siemens investigating WinCC infections, per Symantec	State	Industrial	Confirmed	[30]
2010-07 (undated)	ICS-CERT ICSA-10-201, USB Malware Targeting Siemens Control Software, names SIMATIC WinCC and STEP 7 (the page opened shows only its 2014 revision)	State	Industrial	Confirmed	[22]
2010-07-22	CVE-2010-2568 published; JMicron certificate revoked, per Symantec	Vulnerability	Cross-sector	Confirmed	[27][30]
2010-07-28	FBI, Slovenian and Spanish police announce the arrest of Mariposa's suspected creator	Crime	Consumers, finance	Confirmed	[16]
2010-08-02	Out-of-band MS10-046 fixes CVE-2010-2568, already exploited by a number of malware families	Vulnerability	Cross-sector	Confirmed	[25]
2010-08-06	Stuxnet's PLC code injection reported, per Symantec	State	Industrial	Confirmed	[30]
2010-08-11	CVE-2010-2861, ColdFusion administrator directory traversal, published	Vulnerability	Cross-sector (servers)	Confirmed	[61]
2010-08-22	Iran stops reporting new Stuxnet infections, most likely because outbound connections were blocked, per Symantec	State	Industrial	Confirmed (vendor's analysis)	[30]
2010-08-23	Microsoft Advisory 2269637 on DLL preloading ("binary planting")	Vulnerability class	Technology	Confirmed	[37]
2010-08-31	NISTIR 7628, Guidelines for Smart Grid Cyber Security, final	Standards	Energy	Confirmed	[78]
2010-09-01	Deputy Secretary Lynn's article discloses the 2008 CENTCOM compromise	State	Defense	Confirmed	[38]
2010-09-07	Siemens reports 15 Stuxnet infections worldwide with no adverse impact on the automation system	State	Industrial	Confirmed	[28]

Date	Event	Thread	Sector	Grade	Sources
2010-09-09	CVE-2010-2883 (Reader CoolType.dll) published; exploited in the wild in September	Vulnerability	Cross-sector	Confirmed	[62]
2010-09-14	MS10-061 fixes the Print Spooler flaw CVE-2010-2729	Vulnerability	Cross-sector	Confirmed	[31]
2010-09-17	Microsoft Advisory 2416728, ASP.NET padding oracle CVE-2010-3332	Vulnerability	Web	Confirmed	[39]
2010-09-21	Twitter's "onMouseOver" cross-site scripting incident	Web	Social media	Confirmed	[41]
2010-09-27	Iranian officials say Bushehr staff PCs were infected but the plant was not, as World Nuclear News relays Reuters	State	Nuclear	Reported	[35]
2010-09-27 to 2010-10-01	DHS Cyber Storm III exercise	Exercise	Government, private sector	Confirmed	[79]
2010-09	ICS-CERT ICSA-10-272-01, the Primary Stuxnet Advisory	State	Industrial	Confirmed	[29]
2010-09-28	MS10-070 fixes CVE-2010-3332 under limited, targeted attacks	Vulnerability	Web	Confirmed	[40]
2010-09-30	Symantec's Stuxnet dossier first published; SDNY charges 37 Zeus defendants; Ukraine's SBU detains five	State; Crime	Industrial; finance	Confirmed	[30][42][44]
2010-10-01	FBI announces the disruption of the Zeus theft ring (Operation Trident Breach)	Crime	Finance, municipalities	Confirmed	[43][44]
2010-10-12	October bulletins MS10-071 to MS10-086; MS10-073 fixes CVE-2010-2743, exploited by malware	Vulnerability	Cross-sector	Confirmed	[50][53]
2010-10-19	UK Strategic Defence and Security Review: GBP 650 million for a national cyber programme	Policy	Government	Confirmed	[80]
2010-10-24	Firesheep released	Web	Consumers, web	Confirmed; Demonstrated	[68]
2010-10-27	Mozilla MFSA 2010-73 for CVE-2010-3765, found during an intrusion attempt	Vulnerability	Cross-sector (browsers)	Confirmed	[56]
2010-10-29	CVE-2010-3654 (Flash, Reader authplay) published; exploited in the wild in October	Vulnerability	Cross-sector	Confirmed	[63]
2010-11-03	Microsoft Advisory 2458511, Internet Explorer CVE-2010-3962	Vulnerability	Cross-sector (browsers)	Confirmed	[45]

Date	Event	Thread	Sector	Grade	Sources
2010-11-09	MS10-087 fixes six Office flaws including the RTF flaw CVE-2010-3333	Vulnerability	Cross-sector	Confirmed	[54]
2010-11	USCC 2010 Annual Report to Congress	State; Routing	Government	Confirmed	[10]
2010-11 (late)	WikiLeaks releases State Department cables; PayPal suspends its account, as later charged	Hacktivism	Payments	Confirmed (as charged)	[46]
2010-11-29	President Ahmadinejad says a worm incapacitated some centrifuges, as the Christian Science Monitor reports	State	Nuclear	Reported	[36]
2010-12-03 to 2010-12-14	CVE-2010-4398 (win32k, 3 Dec), CVE-2010-3904 (Linux RDS, 6 Dec) and CVE-2010-4344 (Exim, 14 Dec) published	Vulnerability	Cross-sector	Confirmed	[64]
2010-12-06 to 2010-12-10	Denial-of-service attacks on PayPal ("Operation Avenge Assange"), as charged. Home: Four Books on Cybercrime	Hacktivism	Payments	Confirmed (as charged)	[46]
2010-12-08	US Attorney's release puts Butler's cards at more than USD 87 million in fraud and more than 10,000 banks	Crime	Finance	Confirmed	[14]
2010-12-14	MS10-090 fixes CVE-2010-3962; MS10-092 fixes CVE-2010-3338; MS10-106, the year's last bulletin	Vulnerability	Cross-sector	Confirmed	[45][55][49]
2011-01-18 (boundary)	Two men charged over the AT&T iPad exposure	Exposure	Telecom	Confirmed (as charged)	[20]
2011-02 (boundary)	Symantec's Stuxnet dossier version 1.4; Night Dragon last seen	State	Industrial; energy	Confirmed	[30][47]
2011-07-19 (boundary)	Sixteen arrested in the US over the PayPal attacks and related cases	Hacktivism	Payments	Confirmed (as charged)	[46]
2012 (boundary)	Flame disclosed; ATT&CK dates it to "at least 2010"	State	Middle East	Confirmed	[48]
2021-11-03 to 2026-05-20 (boundary)	KEV adds 2010 CVEs, the last two in 2026	Vulnerability	Cross-sector	Confirmed	[8]

Part 2: the vulnerability record

The Aurora flaw (summary)

The home is Part 4 of the Zero-Day Trade study (Confirmed [87]). Microsoft published Advisory 979352 on 14 January for the Internet Explorer HTML Object Memory Corruption flaw, and version 1.1 on 15 January revised the summary to reflect investigation of limited targeted attacks (Confirmed [3]). CERT/CC published VU#492515 the same day (Confirmed [7]). NVD and the CVE Program published CVE-2010-0249 on 15 January as a use-after-free in Internet Explorer 6, 6 SP1, 7 and 8, "as exploited in the wild in December 2009 and January 2010 during Operation Aurora", with CWE-416 and a CVSS 3.1 score of 8.8 (Confirmed [5][6]). MS10-002, a Critical cumulative update covering Internet Explorer 5.01 to 8, fixed it with seven other CVEs on 21 January; Microsoft said it was aware of limited attacks and thanked Google and Mandiant for details of limited, targeted attacks against Internet Explorer 6 customers (Confirmed [4]). KEV added CVE-2010-0249 on 20 May 2026 (Confirmed [8]).

Exploited at or before the fix

The year's flaws for which the record shows exploitation at or before the fix, in date order:

CVE	Product and flaw	First notice	Exploitation in the record	Fix	CWE (source)	KEV	Sources
CVE-2010-0249	Internet Explorer, HTML object use-after-free	Advisory 979352 and VU#492515, 14 Jan	In the wild Dec 2009 and Jan 2010 during Aurora	MS10-002, 21 Jan	CWE-416 (NVD, KEV)	2026-05-20	[3][7][5][4][8]
CVE-2010-0806	Internet Explorer	Advisory 981374	Microsoft aware of attacks attempting to exploit it	MS10-018, 30 Mar	CWE-399 (KEV)	2026-05-20	[52][8]
CVE-2010-1297	Flash Player, AIR, Reader and Acrobat	CVE published 8 Jun	In the wild in June	Adobe's fix not opened	CWE-787 (KEV)	2022-06-08	[60][8]
CVE-2010-1885	Windows Help and Support Center	Advisory 2219475	Microsoft aware of active attacks	MS10-042, 13 Jul	not checked	not in KEV	[21]
CVE-2010-2568	Windows Shell shortcut icon loading	Advisory 2286198 and US-CERT alert, 16 Jul; VU#940193, 15 Jul	Demonstrated in the wild in July; exploited by a number of malware families at MS10-046's release	MS10-046, 2 Aug (out of band)	CWE-20 (KEV); NVD-CWE-noinfo (NVD)	2022-09-15	[24][23][26][27][25][8]
CVE-2010-2883	Adobe Reader CoolType.dll stack overflow	CVE published 9 Sep	In the wild in September	Adobe's fix not opened	CWE-119 (KEV); CWE-787 (CISA ADP in the CVE record)	2022-06-08	[62][8]
CVE-2010-2729	Windows Print Spooler	MS10-061, 14 Sep	Exploited on a limited scale by malware at release; in the wild in September per the CVE record	MS10-061, 14 Sep	CWE-20 (NVD)	not in KEV	[31][32]
CVE-2010-3332	ASP.NET padding oracle	Advisory 2416728, 17 Sep	Limited, targeted attacks	MS10-070, 28 Sep	not checked	not in KEV	[39][40]

CVE	Product and flaw	First notice	Exploitation in the record	Fix	CWE (source)	KEV	Sources
CVE-2010-2743	Win32k keyboard layout	MS10-073, 12 Oct	Stuxnet demonstrated it in July; exploited on a limited scale by malware at release	MS10-073, 12 Oct	not checked	not in KEV	[32][53]
CVE-2010-3765	Firefox 3.6 on Windows XP	MFSA 2010-73, 27 Oct	Found by Telenor SOC investigating an intrusion attempt; exploited by the Belmoo malware in October	Firefox 3.5.15 and 3.6.12	not checked	2025-10-06	[56][57][8]
CVE-2010-3654	Flash Player and Reader's authplay component	CVE published 29 Oct	In the wild in October	Adobe's fix not opened	not checked	not in KEV	[63]
CVE-2010-3962	Internet Explorer, uninitialized memory	Advisory 2458511, 3 Nov	MS10-090 calls it the vulnerability "currently being exploited"	MS10-090, 14 Dec	not checked	2025-10-06	[45][8]
CVE-2010-3338	Windows Task Scheduler	MS10-092, 14 Dec	Limited, targeted attacks	MS10-092, 14 Dec	CWE-20 (NVD)	not in KEV	[55][32]

Three more flaws are in KEV without an exploitation date in the sources opened here. CVE-2010-0188 in Adobe Reader and Acrobat before 9.3.1 was published on 21 February, and KEV lists it with CWE-94 and flags known ransomware use (Confirmed [58][8]). MS10-087 fixed six Office flaws on 9 November, including the RTF flaw CVE-2010-3333, which NVD published on 10 November with CWE-787 and KEV lists with CWE-119 (Confirmed [54][65][8]). MS10-015 fixed the publicly disclosed kernel flaw CVE-2010-0232 on 9 February with no exploitation known at release; KEV added it on 3 March 2022 (Confirmed [51][8]).

In ATT&CK terms the browser, document and plug-in flaws read as T1203 Exploitation for Client Execution, and the privilege-escalation flaws as T1068 Exploitation for Privilege Escalation; ATT&CK lists T1203 for Elderwood and T1068 for Stuxnet, but tying either to the other 2010 flaws is this study's reading (Confirmed [11][34]; Inferred).

Servers and middleware

Three flaws published in 2010 sat in internet-facing server software and one in the Linux kernel. CVE-2010-0738, published on 28 April, is the JBoss JMX-Console applying access control only to the GET and POST methods (Confirmed [59]). CVE-2010-2861, published on 11 August, is directory traversal in the Adobe ColdFusion administrator console, 9.0.1 and earlier, listed by KEV as CWE-22 (Confirmed [61][8]). CVE-2010-4344, published on 14 December, is a heap overflow in Exim before 4.70, and CVE-2010-3904, published on 6 December, is a Linux RDS local privilege escalation (Confirmed [64]). KEV lists all four, and flags CVE-2010-0738, CVE-2010-1428 (also JBoss), CVE-2010-2861 and CVE-2010-0188 for known ransomware use (Confirmed [8]). The record opened here gives no 2010 exploitation date for the server flaws, so this study does not date their abuse to 2010 (Inferred).

Classes of flaw: DLL preloading and the padding oracle

Microsoft Security Advisory 2269637 of 23 August addressed published research on a remote attack vector for a class of insecure library loading flaws, which it calls binary planting or DLL preloading, caused by applications loading libraries from insufficiently qualified paths (Confirmed [37]). The advisory is about a class, not one product, and gave guidance to developers (Confirmed [37]). ATT&CK's T1574.001 DLL is this study's mapping (Inferred). Advisory 2416728, published 17 September and updated 28 September, covered the ASP.NET padding oracle CVE-2010-3332, and MS10-070 of 28 September fixed it, with Microsoft aware of limited, targeted attacks (Confirmed [39][40]).

Microsoft's year in bulletins

MS10-001 was published on 12 January and MS10-106 on 14 December, and MS10-107 does not exist, so the year carried 106 bulletins by numbering (Confirmed [49]; the count is Inferred). All sixteen October bulletins, MS10-071 to MS10-086, were published on 12 October and carry 49 distinct vulnerability sections, counted from the individual bulletin pages because the monthly summary page would not open (Confirmed [50]; the count is Inferred).

The KEV tail

The KEV catalog, version 2026.11.10, lists 23 CVE-2010 entries (Confirmed [8]; Demonstrated). Their dates added run from 3 November 2021 to 20 May 2026 (Confirmed [8]):

KEV dateAdded	CVE-2010 ids
2021-11-03	CVE-2010-5326
2021-12-10	CVE-2010-1871
2022-03-03	CVE-2010-3333, CVE-2010-0232, CVE-2010-0188
2022-03-25	CVE-2010-4344, CVE-2010-4345, CVE-2010-2861, CVE-2010-3035
2022-03-28	CVE-2010-4398
2022-04-15	CVE-2010-5330
2022-05-25	CVE-2010-0840, CVE-2010-0738, CVE-2010-1428
2022-06-08	CVE-2010-2883, CVE-2010-2572, CVE-2010-1297
2022-09-15	CVE-2010-2568
2023-05-12	CVE-2010-3904
2025-10-06	CVE-2010-3962, CVE-2010-3765
2026-05-20	CVE-2010-0249, CVE-2010-0806

The gap between exploitation and listing runs to 16 years, so a KEV date says when CISA added the flaw to the catalog, not when it was exploited (Inferred [8][5]).

Part 3: Stuxnet in 2010 (summary)

The Stuxnet study is the home, and it already carries ICSA-10-201-01C, ICSA-10-272-01, MS10-046, MS10-061 and the Symantec dossier (Confirmed [87]). Symantec's W32.Stuxnet Dossier, first published on 30 September (version 1.4 of February 2011 opened), dates the year: VirusBlokAda reported the malware on 17 June; Microsoft's Advisory 2286198 and the revocation of the Realtek certificate came on 16 July; Siemens was investigating WinCC infections on 19 July; the JMicron certificate was revoked on 22 July; MS10-046 shipped on 2 August; PLC code injection was reported on 6 August; and MS10-061 shipped on 14 September (Confirmed [30]). Symantec counted about 100,000 infected hosts and more than 40,000 external IP addresses in over 155 countries, roughly 60 percent of infected hosts in Iran; targeted organizations, all with a presence in Iran, were attacked in June and July 2009 and March, April and May 2010; and from 22 August Iran stopped reporting new infections, most likely because outbound connections to the command servers were blocked (Confirmed [30]). Symantec cautions readers against drawing attribution conclusions, and the 2010 primaries opened here name no sponsoring state (Confirmed [30]).

The defender record ran in parallel. CERT/CC's VU#940193 gives 10 July as the date the shortcut flaw became public and 15 July as its first publication (Confirmed [26]). US-CERT's alert of 16 July said that viewing a crafted shortcut's location in Windows Explorer was enough to run code, that AutoRun or AutoPlay on removable drives could trigger it without interaction, and that a malicious website or WebDAV share could also reach it (Confirmed [23]). ICS-CERT's ICSA-10-201-01C said the trojan interacts with Siemens SIMATIC WinCC or STEP 7 and queries SIMATIC databases, reported four infected Siemens customers with no production impact, and recommended disabling shortcut icons, WebClient and AutoRun, least privilege, current antivirus, caution with USB drives and the Siemens update (Confirmed [22]). ICSA-10-238-01B said Stuxnet used five vulnerabilities, MS08-067 and four zero-days, spread by USB, network shares and SQL databases, and carried a kernel-mode rootkit signed with two revoked certificates, and that on 7 September Siemens reported 15 infections worldwide with no adverse impact on the automation system (Confirmed [28]). The Primary Stuxnet Advisory, ICSA-10-272-01, added propagation through STEP 7 project files, WinCC database files and the print spooler flaw fixed by MS10-061, with updates through command and control and peer-to-peer RPC (Confirmed [29]). NVD records CVE-2010-2772, a hard-coded password in Siemens WinCC and PCS 7 (CWE-798), as demonstrated in the wild in July 2010 by Stuxnet (Confirmed [33]). MS10-061 itself does not name Stuxnet (Confirmed [31]).

In Iran, officials said that Bushehr staff computers were infected but the plant was not affected, as World Nuclear News relayed from Reuters on 27 September, and on 29 November President Ahmadinejad said a computer worm had incapacitated some centrifuges, without naming Stuxnet, as the Christian Science Monitor reported (Reported [35][36]). ATT&CK's S0603 Stuxnet was discovered in 2010 with some components used as early as November 2008, is the first publicly reported malware to target ICS devices specifically, and lists no group (Confirmed [34]).

Part 4: the crime record

Max Butler and the 2010 records (summary)

The home is Four Books on Cybercrime (Confirmed [87]). On 12 February Max Ray Butler, known as "Iceman", was sentenced in the Western District of Pennsylvania to 13 years in prison, five years of supervised release and USD 27.5 million in restitution for wire fraud; he hacked financial institutions and card processing centres and co-ran CardersMarket, which had about 4,500 members at its peak, and more than 1.8 million stolen card numbers were found on his systems (Confirmed [13]). A US Attorney's release of 8 December says the cards carried more than USD 87 million in fraudulent charges and affected more than 10,000 US banks (Confirmed [14]).

Gonzalez and the TJX ring (summary)

The home for Heartland and the Gonzalez ring is E-Commerce Threats (Confirmed [87]). A US Attorney's release of 16 April in the District of Massachusetts records that Albert Gonzalez was sentenced on 25 March to 20 years plus three years of supervised release, that co-conspirator Damon Patrick Toey was sentenced to five years and a USD 100,000 fine, and that 40 million distinct card numbers sat on two conspirators' servers (Confirmed [66]). On 29 March Christopher Scott, whose role was wardriving into retailers' wireless networks in Miami from about 2003 to 2007, was sentenced to seven years (Confirmed [67]). The home study also records the 8 January Heartland and Visa settlement of up to USD 60 million for Visa issuers; the SEC exhibit was not reopened for this study (Inferred [82][87]).

Waledac: a civil takedown

On 24 February Microsoft announced "Operation b49": on 22 February a judge in the Eastern District of Virginia had granted a temporary restraining order in Microsoft v. John Does 1-27 (civil action 1:10CV156) that cut off 277 domains of the Waledac spam botnet (Confirmed as the plaintiff's account [15]). Microsoft estimated that Waledac could send more than 1.5 billion spam e-mails a day, that about 651 million Waledac spam e-mails reached Hotmail between 3 and 21 December 2009, and that hundreds of thousands of computers were infected; these are the company's own estimates (Reported [15]). The library's ATT&CK data has no Waledac object (Demonstrated [84]).

Mariposa

The FBI, the Slovenian Criminal Police and the Spanish Guardia Civil announced on 28 July that the Guardia Civil had arrested three suspected Mariposa operators in February and that Slovenian police had arrested the suspected creator, "Iserdo", 23, the week before (Confirmed [16]). The botnet was built with the Butterfly Bot kit, which was sold to hundreds of criminals, and was used to steal passwords and card data and to launch denial-of-service attacks (Confirmed [16]). Industry experts estimated that Mariposa may have infected 8 to 12 million computers; the FBI quotes the figure as an estimate, not a count (Reported [16]).

Zeus and Operation Trident Breach

On 30 September the US Attorney for the Southern District of New York charged 37 defendants in 21 cases over global bank fraud schemes that used the Zeus Trojan and other malware (Confirmed as charged [42]). As charged, Zeus arrived as an apparently benign e-mail at small businesses and municipalities, recorded keystrokes at online-banking login, and the stolen credentials funded unauthorized transfers to hundreds of false-name accounts run by a money mule organization that recruited mules on student visas; more than USD 3 million moved from dozens of compromised accounts (Confirmed as charged [42]). The FBI said on 1 October that Operation Trident Breach had begun in May 2009 when Omaha agents saw ACH payments to 46 accounts, that the ring had stolen USD 70 million and attempted about USD 220 million from medium-sized companies, towns and churches, that partners were in the Netherlands, Ukraine and the UK, that Ukraine's SBU detained five people on 30 September, and that its New York office had arrested 10 people and was seeking 17 others (Confirmed [43][44]). The library's S0016 P2P ZeuS is Gameover ZeuS, a later variant, so this study binds no Zeus software object (Demonstrated [84]).

The AT&T iPad exposure

Per the District of New Jersey complaint unsealed on 18 January 2011, a script that mimicked an iPad 3G guessed ICC-IDs against an AT&T web service from 5 to 9 June 2010 and harvested about 120,000 ICC-ID and e-mail pairs; the data went to Gawker, which published redacted results; a group calling itself Goatse Security was identified; and two men were charged with conspiracy to access AT&T's servers (Confirmed as charged [20]). The complaint as summarized names no software flaw, so this study binds no ATT&CK technique to it (Inferred).

Operation Avenge Assange (summary)

The home is Four Books on Cybercrime (Confirmed [87]). According to a DOJ and FBI release of 19 July 2011, WikiLeaks released State Department cables in late November 2010, PayPal suspended its account, and the San Jose indictment alleges that Anonymous ran denial-of-service attacks on PayPal from 6 to 10 December 2010, which it called "Operation Avenge Assange", with a free open-source tool; 14 people were arrested on that indictment and two more on related complaints, more than 35 search warrants were executed, and UK and Dutch police made one and four arrests (Confirmed as charged [46]).

Part 5: the state and espionage record

Operation Aurora (summary)

The home is the Zero-Day Trade study (Confirmed [87]); the actor view is in China-attributed threat actors. Google's post of 12 January says the attack originated from China and resulted in theft of intellectual property, that at least twenty other large companies were targeted, that a primary goal appeared to be the Gmail accounts of Chinese human rights activists, and that only two accounts appear to have been accessed, limited to account information and subject lines; it does not name the Chinese government, and it advised antivirus, patching and browser updates (Confirmed [1]). On 22 March Google stopped censoring Search, News and Images on Google.cn and redirected users to Google.com.hk, citing the January attacks and the surveillance they uncovered (Confirmed [2]). The Secretary of State's remarks of 21 January said the United States looked to Chinese authorities to conduct a thorough review of the intrusions that led to Google's announcement (Confirmed [9]).

The USCC's 2010 Annual Report says the breach appeared to originate within China and targeted other US companies, and assesses that such efforts likely operate with the tacit knowledge of the Chinese government and may even involve full government support; that is the Commission's assessment, not a finding (Confirmed [10]). The same report quotes iDefense as saying the source IP addresses and servers corresponded to a single foreign entity of Chinese state agents or proxies, says researchers traced the penetration to two schools in China, one with military ties, and, citing Google's information security manager, says the attackers profiled targeted employees on social networks, posed as an acquaintance in a chat session, and got an employee to click a link to a malicious photo-sharing site, reportedly hosted in Taiwan, that downloaded malware (Reported [10]).

ATT&CK describes G0066 Elderwood, aliases Elderwood Gang, Beijing Group and Sneaky Panda, as a suspected Chinese espionage group "reportedly responsible" for the 2009 Google intrusion, and S0203 Hydraq, whose aliases include Aurora and 9002 RAT, as first used by Elderwood in that intrusion and associated with Elderwood and Axiom (G0001) (Confirmed that ATT&CK says so [11][12]; the attribution is Reported). Elderwood's techniques come from reporting across 2009 to 2012 and are not specific to Aurora (Confirmed [11]). The library's Notable Incidents table attributes Aurora to "APT10 / Comment Crew (Chinese PLA Unit 61398, attributed)", which the Hacker and the State study already flags as merging distinct groups; this chronicle uses G0066 as ATT&CK maps it and does not repeat that line (Inferred [87][11]).

Shadows in the Cloud

The Information Warfare Monitor (Citizen Lab and SecDev) and the Shadowserver Foundation published "Shadows in the Cloud: Investigating Cyber Espionage 2.0", JR03-2010, on 6 April, after an eight-month investigation; a Citizen Lab op-ed of 7 April confirmed the release and noted that at least one Canadian institution was ensnared (Confirmed [18][19]). The network compromised government, business and academic systems in India, including its national security establishment, the Offices of the Dalai Lama and the United Nations; the Embassy of Pakistan in the United States was also compromised (Confirmed [18]). The recovered data included Indian government documents, two marked SECRET, six RESTRICTED and five CONFIDENTIAL, possibly taken from personal computers; 1,500 letters sent from the Dalai Lama's office between January and November 2009; and visa applications to Indian missions in Afghanistan from nationals of 13 countries (Confirmed [18]). Command and control used Twitter, Google Groups, Blogspot, Baidu Blogs, blog.com and Yahoo Mail as a top tier, falling back to core servers in the PRC (Confirmed [18]).

On attribution the report says the attackers' identity and motivation remain unknown, places them in Chengdu only through circumstantial evidence, links two Chengdu individuals to the PRC underground hacking community, and recalls that the GhostNet investigation found too little evidence to implicate the Chinese government (Confirmed [18]; the Chengdu location is the researchers' assessment, Reported). ATT&CK has no object for this network, so no group or software id is bound; T1102 Web Service is this study's mapping of the layered command and control (no ATT&CK object: Demonstrated [84]; the T1102 mapping: Inferred from [18]).

The China Telecom route event

The USCC's 2010 Annual Report says that for about 18 minutes on 8 April China Telecom advertised erroneous routes that sent traffic for about 15 percent of internet destinations through Chinese servers, affecting .gov and .mil sites, among them the Senate, the Army, Navy, Marine Corps and Air Force, the Office of the Secretary of Defense, NASA, Commerce and NOAA, and commercial sites including Dell, Yahoo, Microsoft and IBM (Confirmed [10]). It says the erroneous data appear to have originated at IDC China Telecommunication and that it has no way to determine what, if anything, Chinese telecommunications firms did with the traffic (Confirmed [10]). The report makes no finding of intent, and no ATT&CK object or technique binds cleanly, so this study binds none (Confirmed [10]; Inferred).

Night Dragon

ATT&CK's campaign C0002 Night Dragon targeted oil, energy and petrochemical companies and individuals in Kazakhstan, Taiwan, Greece and the United States, first seen in November 2009 and last seen in February 2011, so 2010 is its only full year (Confirmed [47]). The actors are unidentified and researchers assessed a China-based group; ATT&CK lists 29 techniques and the software ASPXSpy (S0073), at (S0110), gsecdump (S0008), PsExec (S0029) and zwShell (S0350) (Confirmed [47]; Demonstrated [84]). The China-attributed threat actors study carries Night Dragon in its campaign table (Confirmed [87]).

Buckshot Yankee, disclosed

Deputy Secretary of Defense William J. Lynn III wrote in Foreign Affairs, published on 1 September, that in 2008 an infected flash drive inserted into a US military laptop at a base in the Middle East, carrying code placed by a foreign intelligence agency, spread across Central Command's classified and unclassified systems (Confirmed [38]). He called it the most significant breach of US military computers ever and named the response Operation Buckshot Yankee (Confirmed [38]). The event is 2008; 2010 is the disclosure (Confirmed [38]).

Written later: Flame and Operation Dust Storm

ATT&CK's S0143 Flame is an information-collection toolkit active since at least 2010 that primarily targeted Middle Eastern countries, with no group listed; it was disclosed in 2012 (Confirmed [48]); the United States-attributed threat actors study carries its technical record. The library's ATT&CK data gives campaign C0016 Operation Dust Storm a first-seen date of January 2010 and a last-seen date of February 2016 (Demonstrated [84]). Neither was known in 2010, so neither is in this study's Navigator layer (Inferred).

Part 6: the web, trust and defaults

Apache's JIRA attack

The Apache Software Foundation reported on 13 April a direct, targeted attack on its JIRA host, brutus.apache.org, from 5 to 9 April (Confirmed [17]). As Apache describes it, a cross-site scripting link posted in a new issue stole administrators' sessions; a parallel brute-force attack on the login page succeeded on 6 April; the attackers changed the attachment storage path so that JSP files would run; they copied home directories; and by 9 April they had installed a component that captured passwords (Confirmed [17]). The hashed passwords of JIRA, Bugzilla and Confluence users were compromised (Confirmed [17]). In ATT&CK terms this is T1539 Steal Web Session Cookie and T1110 Brute Force (basis Described), with T1505.003 Web Shell for the JSP step (basis Inferred); the mapping is this study's (Inferred from [17]).

Twitter's onMouseOver incident

Twitter wrote on 21 September that it was notified at 2:54 am PDT of a cross-site scripting exploit that had surfaced about half an hour earlier, and that it fixed the primary issue by 7:00 am PDT (Confirmed [41]). The flaw had been patched the month before and was resurfaced by an unrelated site update (Confirmed [41]). CWE-79 and CAPEC-63 are this study's pairing (Inferred).

Firesheep and encryption by default

On 12 January Gmail began turning HTTPS on by default for all users; the option had existed since 2008 (Confirmed [76]). On 17 June EFF and the Tor Project released HTTPS Everywhere, a Firefox extension that moved connections to sites such as Google Search, Wikipedia, Twitter and Facebook onto HTTPS (Confirmed [77]). On 24 October Eric Butler released Firesheep, which showed that anyone on an open wireless network could capture unencrypted session cookies and take over nearby users' accounts ("sidejacking"); its author argued that only full end-to-end HTTPS fixes this (Confirmed as the author's post [68]; Demonstrated by the tool). T1539 Steal Web Session Cookie, T1550.004 Web Session Cookie and T1040 Network Sniffing are this study's mapping; ATT&CK does not cite Firesheep (Inferred).

The protocol and the root

RFC 5746, the TLS Renegotiation Indication Extension, was published as a Proposed Standard in February; it binds each renegotiation to its enclosing connection so an attacker cannot prefix traffic to a client's session (Confirmed [70]). The first DNS root trust anchor, KSK-2010 with key tag 19036, is valid from 15 July 2010 to 11 January 2019 in IANA's root-anchors file, which marks when the signed root became verifiable (Confirmed [73]).

Paying for bugs

The Chromium project announced a vulnerability reward program on 28 January, with a USD 500 base reward and USD 1,337 for especially severe or clever bugs, crediting Mozilla's long-running program (Confirmed [74]). Mozilla raised its bounty to USD 3,000 per eligible bug for reports from 1 July, announced on 15 July, and added Firefox Mobile and the Mozilla services those products rely on (Confirmed [75]). The Zero-Day Trade study reads Google's reward programs as following Aurora; that reading is the home study's (Inferred [87]).

Part 7: law, policy and standards

Organizing for cyber

Deputy Secretary Lynn wrote that US Cyber Command, ordered by Secretary Gates in June 2009, began operations in May 2010 under US Strategic Command, was slated to be fully operational by October, and would integrate cyberdefense operations (Confirmed [38]). DHS's Cyber Storm III ran from 27 September to 1 October with 8 Cabinet-level departments, 13 states, 12 international partners and about 60 private-sector companies; it tested the National Cyber Incident Response Plan and the NCCIC, and no actual networks were targeted (Confirmed [79]). The UK's Strategic Defence and Security Review, published on 19 October, set up a transformative national programme to protect the UK in cyberspace backed by GBP 650 million of new funds (Confirmed [80]). The Obama White House archive publishes a summary page for the CNCI, the federal cybersecurity initiative, starting with Initiative 1, Trusted Internet Connections; the page carries no date, and its 2010 declassification date was not checked (Confirmed [81]; not checked).

Standards and frameworks

NIST published SP 800-37 Revision 1, Guide for Applying the Risk Management Framework to Federal Information Systems: A Security Life Cycle Approach, in February 2010; it was updated on 5 June 2014, withdrawn on 20 December 2019 and superseded by Revision 2 (Confirmed [69]). NIST published NISTIR 7628, Guidelines for Smart Grid Cyber Security, in final form on 31 August 2010; it was withdrawn on 25 September 2014 and superseded by Revision 1 (Confirmed [78]). The 2010 CWE/SANS Top 25 is dated 29 March, and its top five are CWE-79 (cross-site scripting), CWE-89 (SQL injection), CWE-120 (classic buffer overflow), CWE-352 (cross-site request forgery) and CWE-285 (improper authorization) (Confirmed [71]). The OWASP Top 10 2010 puts Injection at A1 and cross-site scripting at A2, adds Security Misconfiguration at A6, debuts Unvalidated Redirects and Forwards at A10, and places Insufficient Transport Layer Protection at A9; the document says the Top 10 was first released in 2003 with minor updates in 2004 and 2007 (Confirmed [72]). Its release day, 19 April, is taken from the PDF's creation metadata and is approximate (Confirmed [72]).

Part 8: the defender view

Removable media needed a policy, not a default. The shortcut flaw ran code from a drive viewed in a file manager, and AutoRun or AutoPlay could trigger it without interaction (Confirmed [23][22]). Disabling AutoRun and restricting removable media are ATT&CK mitigations in substance: M1042 Disable or Remove Feature or Program and M1034 Limit Hardware Installation (Inferred).

Workarounds bought time. The shortcut flaw had no fix for at least the 17 days between Advisory 2286198 and MS10-046; the workarounds MS10-046 records were to disable shortcut icon display and the WebClient service and to block LNK and PIF files at the gateway and outbound SMB at the perimeter, and ICS-CERT added disabling AutoRun (Confirmed [25][24][22]; Inferred for the count). The vulnerability management reference covers working from advisories as well as bulletins (Inferred).

Signed was not safe. Stuxnet's drivers were signed with two certificates that had to be revoked in July (Confirmed [28][30]). Certificate Analysis is the matching D3FEND page; the library links no countermeasure to T1553.002 (Demonstrated [85]; Inferred for the pairing).

Default credentials were an industrial weakness. The hard-coded WinCC password was in the field when Stuxnet used it (Confirmed [33]). Change Default Password is the countermeasure the library links to T1078.001 (Demonstrated [85]).

Encryption had to cover the session, not just the login. Firesheep took cookies from sites that encrypted only the login, and Gmail and HTTPS Everywhere moved whole sessions to HTTPS (Confirmed [68][76][77]). M1041 Encrypt Sensitive Information and Message Encryption are the matching controls (Inferred).

Fraud detection needed the bank's side. Zeus captured credentials at the customer's keyboard and the theft surfaced as ACH payments to mule accounts, which is how Omaha agents first saw it (Confirmed [42][44]). M1032 Multi-factor Authentication and M1017 User Training are the matching mitigations (Inferred).

Takedown was a legal tool. Waledac was cut off by a civil restraining order, and Mariposa's suspected operators and creator were arrested in two countries (Confirmed [15][16]). The incident response reference carries the coordination side (Inferred).

Collaboration tools were targets. Apache's issue tracker gave the attackers a cross-site scripting link and a login page to brute-force, and an admin path to run code (Confirmed [17]). M1036 Account Use Policies,

M1032 Multi-factor Authentication and M1048 Application Isolation and Sandboxing are the matching mitigations (Inferred).

Part 9: MITRE mappings

No 2010 record opened here carries an ATT&CK id; every binding below is a later mapping (Inferred [5][11]). Every Enterprise binding is either a technique ATT&CK lists for an object it ties to 2010 (G0066 Elderwood, S0203 Hydraq, C0002 Night Dragon, S0603 Stuxnet) or this study's mapping of a behavior a source describes (Inferred). Every id was checked against the library's data files (`data/attack/technique_profiles.jsonl`, `software_to_technique.jsonl`, `group_to_technique.jsonl`, `campaign_profiles.jsonl`, `groups.jsonl`, `software.jsonl`, `technique_to_d3fend.jsonl`, `data/attack/ics/` and `data/weaknesses/`) and against the generated pages under `mitre/` on 12 and 13 November 2026 (Demonstrated [84][85][86]). Names and tactics are those of ATT&CK v20.0 as the library carries them, including the v20.0 tactics stealth and defense-impairment (Demonstrated [84]).

The Basis column says how strong each tie is, as in the 2016 chronicle. **Listed** means ATT&CK itself lists the id for the object. **Described** means a primary source describes the behavior and this study assigns the id. **Reported** means only a Reported source describes it. **Inferred** means the mapping rests on this study's reading. Two caveats apply to the Listed rows: Elderwood's and Hydraq's tie to Aurora is ATT&CK's "reportedly" and "first used", with techniques drawn from reporting across 2009 to 2012, and only the Stuxnet techniques that a 2010 primary describes are listed here; the Stuxnet study carries all 44 (Confirmed [11][12][34]; Demonstrated [84]).

ATT&CK Enterprise techniques

Id	Name (ATT&CK v20.0)	Tactic(s)	Basis	Dated event or report	Sources
T1189	Drive-by Compromise	initial-access	Listed (G0066)	Elderwood, reportedly Aurora (disclosed 12 Jan); the USCC's account of a link to a malicious photo-sharing site (Nov; Reported)	[11][10]
T1203	Exploitation for Client Execution	execution	Listed (G0066); Described	CVE-2010-0249 in limited, targeted attacks (MS10-002, 21 Jan); Internet Explorer, Firefox, Reader and Flash flaws exploited at or before their fixes (Jan to Dec; match Inferred)	[11][4][52][56][60][62][63][45]
T1105	Ingress Tool Transfer	command-and-control	Listed (G0066, S0203, C0002); Described	Elderwood, Hydraq, Night Dragon; the Aurora link downloaded malware (USCC; Reported)	[11][12][47][10]
T1027.002	Software Packing	stealth	Listed (G0066, C0002)	Elderwood; Night Dragon (ATT&CK objects)	[11][47]
T1027.013	Encrypted/Encoded File	stealth	Listed (G0066, C0002)	Elderwood; Night Dragon (ATT&CK objects)	[11][47]
T1566.001	Spearphishing Attachment	initial-access	Listed (G0066)	Elderwood (ATT&CK object)	[11]
T1566.002	Spearphishing Link	initial-access	Listed (G0066, C0002)	Elderwood; Night Dragon	[11][47]
T1204.001	Malicious Link	execution	Listed (G0066, C0002)	Elderwood; Night Dragon; the Aurora employee clicked a link (USCC; Reported)	[11][47][10]
T1204.002	Malicious File	execution	Listed (G0066)	Elderwood (ATT&CK object)	[11]
T1005	Data from Local System	collection	Listed (S0203, C0002); Described	Hydraq; Night Dragon; Apache attackers copied home directories (5 to 9 Apr)	[12][47][17]
T1007	System Service Discovery	discovery	Listed (S0203)	Hydraq (ATT&CK object)	[12]
T1012	Query Registry	discovery	Listed (S0203)	Hydraq (ATT&CK object)	[12]

Id	Name (ATT&CK v20.0)	Tactic(s)	Basis	Dated event or report	Sources
T1016	System Network Configuration Discovery	discovery	Listed (S0203)	Hydraq (ATT&CK object)	[12]
T1027	Obfuscated Files or Information	stealth	Listed (S0203)	Hydraq (ATT&CK object)	[12]
T1048	Exfiltration Over Alternative Protocol	exfiltration	Listed (S0203)	Hydraq (ATT&CK object)	[12]
T1057	Process Discovery	discovery	Listed (S0203)	Hydraq (ATT&CK object)	[12]
T1070.004	File Deletion	stealth	Listed (S0203)	Hydraq (ATT&CK object)	[12]
T1082	System Information Discovery	discovery	Listed (S0203)	Hydraq (ATT&CK object)	[12]
T1083	File and Directory Discovery	discovery	Listed (S0203, C0002)	Hydraq; Night Dragon	[12][47]
T1112	Modify Registry	defense-impairment, persistence	Listed (S0203, C0002)	Hydraq; Night Dragon	[12][47]
T1113	Screen Capture	collection	Listed (S0203)	Hydraq (ATT&CK object)	[12]
T1129	Shared Modules	execution	Listed (S0203)	Hydraq (ATT&CK object)	[12]
T1134	Access Token Manipulation	stealth, privilege-escalation	Listed (S0203)	Hydraq (ATT&CK object)	[12]
T1543.003	Windows Service	persistence, privilege-escalation	Listed (S0203)	Hydraq (ATT&CK object)	[12]
T1569.002	Service Execution	execution	Listed (S0203)	Hydraq (ATT&CK object)	[12]
T1573.001	Symmetric Cryptography	command-and-control	Listed (S0203)	Hydraq (ATT&CK object)	[12]
T1685.005	Clear Windows Event Logs	defense-impairment	Listed (S0203)	Hydraq (ATT&CK object)	[12]
T1003.002	Security Account Manager	credential-access	Listed (C0002)	Night Dragon (Nov 2009 to Feb 2011)	[47]
T1008	Fallback Channels	command-and-control	Listed (C0002)	Night Dragon	[47]
T1033	System Owner/User Discovery	discovery	Listed (C0002)	Night Dragon	[47]
T1059.003	Windows Command Shell	execution	Listed (C0002)	Night Dragon	[47]

Id	Name (ATT&CK v20.0)	Tactic(s)	Basis	Dated event or report	Sources
T1071.001	Web Protocols	command-and-control	Listed (C0002)	Night Dragon	[47]
T1074.002	Remote Data Staging	collection	Listed (C0002)	Night Dragon	[47]
T1078	Valid Accounts	stealth, persistence, privilege-escalation, initial-access	Listed (C0002)	Night Dragon	[47]
T1078.002	Domain Accounts	stealth, persistence, privilege-escalation, initial-access	Listed (C0002)	Night Dragon	[47]
T1110.002	Password Cracking	credential-access	Listed (C0002)	Night Dragon	[47]
T1114.001	Local Email Collection	collection	Listed (C0002)	Night Dragon	[47]
T1133	External Remote Services	persistence, initial-access	Listed (C0002)	Night Dragon	[47]
T1190	Exploit Public-Facing Application	initial-access	Listed (C0002); Described	Night Dragon; ASP.NET padding oracle CVE-2010-3332 under limited, targeted attacks (Sep); JBoss, ColdFusion and Exim flaws in KEV (exploitation date not opened; match Inferred)	[47][39][40][8]
T1219	Remote Access Tools	command-and-control	Listed (C0002)	Night Dragon	[47]
T1550.002	Pass the Hash	lateral-movement	Listed (C0002)	Night Dragon	[47]
T1568	Dynamic Resolution	command-and-control	Listed (C0002)	Night Dragon	[47]
T1583.004	Server	resource-development	Listed (C0002)	Night Dragon	[47]
T1584.004	Server	resource-development	Listed (C0002)	Night Dragon	[47]
T1588.001	Malware	resource-development	Listed (C0002)	Night Dragon	[47]
T1588.002	Tool	resource-development	Listed (C0002)	Night Dragon	[47]
T1608.001	Upload Malware	resource-development	Listed (C0002)	Night Dragon	[47]
T1685	Disable or Modify Tools	defense-impairment	Listed (C0002)	Night Dragon	[47]

Id	Name (ATT&CK v20.0)	Tactic(s)	Basis	Dated event or report	Sources
T1091	Replication Through Removable Media	lateral-movement, initial-access	Listed (S0603); Described	Stuxnet's shortcut flaw from USB drives (Jul); the 2008 CENTCOM flash drive (disclosed 1 Sep)	[34][22][23][25][38]
T1210	Exploitation of Remote Services	lateral-movement	Listed (S0603); Described	Stuxnet through MS08-067 and the print spooler flaw (MS10-061, 14 Sep; ICSA-10-272-01)	[34][29][31]
T1021.002	SMB/Windows Admin Shares	lateral-movement	Listed (S0603); Described	Stuxnet through network shares (ICSA-10-238-01B, ICSA-10-272-01)	[34][28][29]
T1080	Taint Shared Content	lateral-movement	Listed (S0603); Described	Stuxnet through STEP 7 project files (ICSA-10-272-01; match Inferred)	[34][29]
T1505.001	SQL Stored Procedures	persistence	Listed (S0603); Described	Stuxnet through WinCC database files and SQL databases (ICSA-10-238-01B, ICSA-10-272-01; match Inferred)	[34][28][29]
T1068	Exploitation for Privilege Escalation	privilege-escalation	Listed (S0603); Described	Stuxnet's privilege-escalation zero-days CVE-2010-2743 and CVE-2010-3338 (MS10-073, MS10-092); CVE-2010-3904 in KEV (match Inferred)	[34][32][53][55][64]
T1014	Rootkit	stealth	Listed (S0603); Described	Stuxnet's signed kernel-mode rootkit (ICSA-10-238-01B)	[34][28]
T1553.002	Code Signing	defense-impairment	Listed (S0603); Described	Drivers signed with the Realtek and JMicron certificates, revoked 16 and 22 Jul	[34][28][30]
T1078.001	Default Accounts	stealth, persistence, privilege-escalation, initial-access	Listed (S0603); Described	Hard-coded WinCC and PCS 7 password CVE-2010-2772 (Jul)	[34][33]
T1570	Lateral Tool Transfer	lateral-movement	Listed (S0603)	Stuxnet's peer-to-peer RPC updates (ICSA-10-272-01; match Inferred)	[34][29]

Id	Name (ATT&CK v20.0)	Tactic(s)	Basis	Dated event or report	Sources
T1102	Web Service	command-and-control	Described	Shadows in the Cloud: command and control over Twitter, Google Groups, Blogspot, Baidu Blogs, blog.com and Yahoo Mail (6 Apr)	[18]
T1539	Steal Web Session Cookie	credential-access	Described	Apache JIRA cross-site scripting link stole administrators' sessions (5 to 9 Apr); Firesheep (24 Oct; match Inferred)	[17][68]
T1110	Brute Force	credential-access	Described	Apache JIRA login brute-forced, succeeding on 6 Apr	[17]
T1566	Phishing	initial-access	Described	Zeus sent as an apparently benign e-mail (charged 30 Sep); the release does not say attachment or link	[42]
T1056.001	Keylogging	collection, credential-access	Described	Zeus recorded keystrokes at online-banking login (charged 30 Sep)	[42]
T1657	Financial Theft	impact	Described	Zeus transfers to mule accounts (30 Sep); Trident Breach's USD 70 million (1 Oct)	[42][43][44]
T1498	Network Denial of Service	impact	Described	PayPal attacks (6 to 10 Dec, as charged); Mariposa used for denial-of-service attacks (Jul)	[46][16]
T1593.001	Social Media	reconnaissance	Reported	Aurora attackers profiled employees on social networks (USCC citing Google's security manager)	[10]
T1566.003	Spearphishing via Service	initial-access	Reported	Aurora attackers posed as an acquaintance in a chat session (USCC citing Google's security manager)	[10]
T1505.003	Web Shell	persistence	Inferred	Apache attackers changed the attachment path so JSP files would run (Apr)	[17]

Id	Name (ATT&CK v20.0)	Tactic(s)	Basis	Dated event or report	Sources
T1550.004	Web Session Cookie	lateral-movement	Inferred	Firesheep account takeover with captured cookies (24 Oct); Apache's stolen administrator sessions (Apr)	[68][17]
T1040	Network Sniffing	credential-access, discovery	Inferred	Firesheep captured cookies on open Wi-Fi (24 Oct)	[68]
T1574.001	DLL	stealth, execution	Inferred	DLL preloading ("binary planting") class, Advisory 2269637 (23 Aug); no exploited event in the record	[37]

Left out on purpose: any technique for the China Telecom route event, which the USCC says it cannot interpret [10]; any technique for the AT&T iPad exposure, whose complaint as summarized names no flaw [20]; any technique for Waledac beyond its spam, which ATT&CK does not model as a technique, and any Waledac, Mariposa or Zeus software object, none of which the library's data carries for 2010 (S0016 P2P ZeuS is the later Gameover variant) [15][16][42][84]; T1189 for CVE-2010-1885 and the other browser flaws, whose delivery the bulletins opened here do not describe [21]; T1557 Adversary-in-the-Middle for RFC 5746, which fixed a protocol gap with no exploitation in the record [70]; the 30 Stuxnet techniques no 2010 primary opened here describes, which the home study binds [34][87]; and Flame's and Operation Dust Storm's techniques, which were written years later [48][84].

ATT&CK for ICS

Stuxnet is the year's industrial event, and the Stuxnet study carries its full ICS mapping, 25 techniques in the library's ICS data (Confirmed [87]; Demonstrated [84]). The table lists the ten ICS techniques ATT&CK lists for S0603 that the 2010 record most directly supports; names and tactics are those in `data/attack/ics/`. The library has no generated pages for ICS techniques, so the ids are plain text; the ten are saved in the study's ATT&CK for ICS layer, described under Navigator layers below (Demonstrated [84][86]).

Id	Name (ATT&CK for ICS v20.0)	Tactic(s)	Object	Basis	Dated event or report	Sources
T0847	Replication Through Removable Media	initial-access	S0603 Stuxnet	Listed; Described	USB malware targeting Siemens control software (Jul)	[34][22]
T0866	Exploitation of Remote Services	initial-access, lateral-movement	S0603 Stuxnet	Listed; Described	MS08-067 and the print spooler flaw (Sep)	[34][29]
T0873.001	Siemens Project File Format	persistence	S0603 Stuxnet	Listed; Described	Propagation through STEP 7 project files (ICSA-10-272-01)	[34][29]
T0843	Program Download	lateral-movement	S0603 Stuxnet	Listed; Described	PLC code injection reported (6 Aug)	[34][30]
T0889	Modify Program	persistence	S0603 Stuxnet	Listed	Stuxnet (ATT&CK object)	[34]

Id	Name (ATT&CK for ICS v20.0)	Tactic(s)	Object	Basis	Dated event or report	Sources
T0821	Modify Controller Tasking	execution	S0603 Stuxnet	Listed	Stuxnet (ATT&CK object)	[34]
T0851	Rootkit	evasion, inhibit-response-function	S0603 Stuxnet	Listed	Stuxnet (ATT&CK object)	[34]
T0836	Modify Parameter	impair-process-control	S0603 Stuxnet	Listed	Stuxnet (ATT&CK object)	[34]
T0831	Manipulation of Control	impact	S0603 Stuxnet	Listed	Stuxnet (ATT&CK object)	[34]
T0832	Manipulation of View	impact	S0603 Stuxnet	Listed	Stuxnet (ATT&CK object)	[34]

ATT&CK also lists T0882 Theft of Operational Information for S0143 Flame; it is context only, because Flame was disclosed in 2012 (Confirmed [48]; Demonstrated [84]).

ATT&CK groups, software and campaigns

Object	Id	Binding	Sources
Elderwood	G0066	Bound with a caveat: ATT&CK's "suspected" group "reportedly responsible" for Aurora; Google names no state; all 9 techniques in the layer	[11][1]
Hydraq	S0203	Bound with a caveat: ATT&CK says first used by Elderwood in the 2009 Google intrusion; aliases include Aurora; all 19 techniques in the layer	[12]
Axiom	G0001	Context: ATT&CK associates Hydraq with Axiom as well as Elderwood	[12]
PoisonIvy, Briba, Naid, Wiarp, Vasport, Pasam, Nerex, Linfo	S0012, S0204, S0205, S0206, S0207, S0208, S0210, S0211	Context: listed among Elderwood's software across 2009 to 2012 reporting; not tied to a 2010 event here	[11][84]
Night Dragon	C0002	Bound: active through all of 2010; all 29 techniques in the layer	[47]
ASPXSpy, at, gsecdump, PsExec, zwShell	S0073, S0110, S0008, S0029, S0350	Bound through C0002, whose software ATT&CK lists	[47][84]
Stuxnet	S0603	Bound: discovered 2010; 10 of its 44 Enterprise techniques and 10 of its 25 ICS techniques bound here, the rest in the home study	[34][84]
Flame	S0143	Context only: "active since at least 2010", disclosed 2012	[48]
Operation Dust Storm	C0016	Context only: first seen January 2010 in the library's data	[84]

The library's ATT&CK data has no object for Waledac, Mariposa or the Butterfly Bot, the 2010 Zeus ring, the Shadows in the Cloud operators, the PayPal attackers or the Apache attackers (Demonstrated [84]).

CWE and CAPEC

The CWE column names the source of each id. Names are those in the library's CWE and CAPEC data, and every CAPEC listed has a page under `/mitre/capec/` (Demonstrated [85][86]). The library has no per-CWE generated pages, so CWE ids are plain text; the CWE reference is the library's entry point. The CAPEC column uses the ids the library's CWE, CAPEC or technique data relates, unless marked as this study's pairing (Demonstrated [85]).

CWE	Weakness (library name)	Basis	2010 record	CAPEC
CWE-416	Use After Free	NVD, KEV	Internet Explorer (CVE-2010-0249, Aurora)	None (the library relates none)
CWE-787	Out-of-bounds Write	KEV; NVD; CISA ADP	Flash and Reader (CVE-2010-1297, KEV); Office RTF (CVE-2010-3333, NVD); Reader CoolType (CVE-2010-2883, ADP)	None (the library relates none)
CWE-119	Improper Restriction of Operations within the Bounds of a Memory Buffer	KEV	Reader CoolType (CVE-2010-2883); Office RTF (CVE-2010-3333)	CAPEC-100 Overflow Buffers
CWE-94	Improper Control of Generation of Code	KEV	Reader and Acrobat (CVE-2010-0188)	None assigned (the library relates CAPEC-242, CAPEC-35 and CAPEC-77; none is specific to this flaw)
CWE-20	Improper Input Validation	KEV; NVD	Shortcut flaw (CVE-2010-2568, KEV); Print Spooler (CVE-2010-2729, NVD); Task Scheduler (CVE-2010-3338, NVD)	None assigned
CWE-22	Improper Limitation of a Pathname to a Restricted Directory	KEV	ColdFusion administrator (CVE-2010-2861)	CAPEC-126 Path Traversal, which the library relates
CWE-798	Use of Hard-coded Credentials	NVD	Siemens WinCC and PCS 7 (CVE-2010-2772, Stuxnet)	CAPEC-70 Try Common or Default Usernames and Passwords; CAPEC-191 Read Sensitive Constants Within an Executable
NVD-CWE-noinfo	(as NVD lists it)	NVD	Shortcut flaw (CVE-2010-2568)	None
CWE-319	Cleartext Transmission of Sensitive Information	Study's pairing	Firesheep's cleartext session cookies (24 Oct)	CAPEC-102 Session Sidejacking; CAPEC-65 Sniff Application Code
CWE-79	Improper Neutralization of Input During Web Page Generation	Top 25 entry; study's pairing for events	2010 Top 25 rank 1; Apache JIRA link (Apr) and Twitter onMouseOver (21 Sep)	CAPEC-63 Cross-Site Scripting (XSS)
CWE-427	Uncontrolled Search Path Element	Study's pairing	DLL preloading, Advisory 2269637 (23 Aug)	CAPEC-471 Search Order Hijacking, which the library relates

CWE	Weakness (library name)	Basis	2010 record	CAPEC
CWE-89, CWE-120, CWE-352, CWE-285	Improper Neutralization of Special Elements used in an SQL Command; Buffer Copy without Checking Size of Input; Cross-Site Request Forgery; Improper Authorization	Top 25 entries, no 2010 event bound	2010 Top 25 ranks 2 to 5	CAPEC-66 SQL Injection; CAPEC-100 Overflow Buffers; CAPEC-62 Cross Site Request Forgery
none bound	(removable media)	Library data	Stuxnet's USB spread; the CENTCOM flash drive	CAPEC-457 USB Memory Attacks (the library lists it on T1091)
none bound	(code signing)	Library data	Stuxnet's signed drivers	CAPEC-68 Subvert Code-signing Facilities (on T1553.002)
none bound	(rootkit)	Library data	Stuxnet's kernel rootkit	CAPEC-552 Install Rootkit (on T1014)
none bound	(shared content)	Library data	STEP 7 project files	CAPEC-562 Modify Shared File (on T1080)
none bound	(admin shares)	Library data	Stuxnet through network shares	CAPEC-561 Windows Admin Shares with Stolen Credentials (on T1021.002)
none bound	(session theft)	Library data	Apache JIRA; Firesheep	CAPEC-21 Exploitation of Trusted Identifiers; CAPEC-31 Accessing/Intercepting/Modifying HTTP Cookies (on T1539)
none bound	(brute force)	Library data	Apache JIRA login	CAPEC-112 Brute Force (on T1110)
none bound	(web shell)	Library data	Apache JSP step	CAPEC-650 Upload a Web Shell to a Web Server (on T1505.003)
none bound	(keylogging)	Library data	Zeus at banking login	CAPEC-568 Capture Credentials via Keylogger (on T1056.001)
none bound	(phishing)	Library data	Zeus by e-mail; Aurora chat lure; Elderwood; Night Dragon	CAPEC-98 Phishing (on T1566); CAPEC-163 Spear Phishing (on T1566.001, T1566.002 and T1566.003)
none bound	(Night Dragon credentials)	Library data	Night Dragon	CAPEC-55 Rainbow Table Password Cracking (on T1110.002); CAPEC-644 Use of Captured Hashes (Pass The Hash) (on T1550.002); CAPEC-555 Remote Services with Stolen Credentials (on T1133); CAPEC-560 Use of Known Domain Credentials (on T1078)
none bound	(library loading)	Library data	DLL preloading advisory	CAPEC-471 Search Order Hijacking; CAPEC-641 DLL Side-Loading (on T1574.001)
none bound	(protocol, no event)	Library data	RFC 5746 context	CAPEC-94 Adversary in the Middle (AiTM) (on T1557; no 2010 event bound)

D3FEND countermeasures

The technique-to-countermeasure links come from the library's D3FEND data; pairing them with the 2010 record is this study's reading (Demonstrated [85]; Inferred). Every page exists under `/mitre/d3fend/` (Demonstrated [86]).

Technique or event from the record	D3FEND countermeasures (D3FEND tactic)	2010 analogue
T1203, T1068	Process Segment Execution Prevention (Harden); Segment Address Offset Randomization (Harden); Stack Frame Canary Validation (Harden); Shadow Stack Comparisons (Detect); Memory Boundary Tracking (Detect); Process Code Segment Verification (Detect)	The exploited browser, document and kernel flaws [4][45][53][55]; no 2010 source opened here names these controls (Inferred)
T1091	IO Port Restriction (Isolate); Hardware Component Inventory (Model)	Stuxnet's USB spread and ICS-CERT's USB advice [22]; the CENTCOM flash drive [38]
T1189	URL Analysis (Detect); URL Reputation Analysis (Detect); Network Traffic Filtering (Isolate)	The Aurora link to a malicious site (Reported [10])
T1210	Network Traffic Filtering (Isolate); Connection Attempt Analysis (Detect)	Stuxnet through MS08-067 and the print spooler [29][31]; MS10-046's advice to block outbound SMB [25]
T1078.001	Change Default Password (Harden); Account Locking (Evict)	The hard-coded WinCC password [33]
T1014	File Integrity Monitoring (Detect); Software Update (Harden)	Stuxnet's rootkit [28]
T1553.002 (no link in the library)	Certificate Analysis (study's pairing)	Revoked Realtek and JMicron certificates [30] (Inferred)
T1080	Network Resource Access Mediation (Isolate); Decoy Network Resource (Deceive)	STEP 7 project files on shares [29]
T1566.001	Email Filtering (Isolate); File Analysis (Detect); Dynamic Analysis (Detect)	Elderwood's attachments [11]; Zeus by e-mail [42] (Inferred)
T1056.001	Input Device Analysis (Detect)	Zeus keylogging at banking login [42]
T1539, T1550.004	Credential Revocation (Evict); Multi-factor Authentication (Harden); Authentication Cache Invalidation (Evict)	Apache's stolen sessions [17]; Firesheep [68]
T1190	Inbound Traffic Filtering (Isolate); Client-server Payload Profiling (Detect)	ASP.NET padding oracle [39]; JBoss, ColdFusion and Exim in KEV [8]; Night Dragon [47]
T1505.003	Executable Allowlisting (Isolate); File Integrity Monitoring (Detect); Process Spawn Analysis (Detect)	Apache's JSP step [17]
T1102	Outbound Traffic Filtering (Isolate); Relay Pattern Analysis (Detect)	Shadows in the Cloud's command and control over web services [18]

Technique or event from the record	D3FEND countermeasures (D3FEND tactic)	2010 analogue
T1574.001	File Integrity Monitoring (Detect); Local File Permissions (Isolate)	DLL preloading advisory [37]
T1040	DNS Traffic Analysis (Detect), the only countermeasure the library links	Firesheep [68]; the closer 2010 control was Message Encryption (study's pairing, Inferred)

The library links T1110, T1566, T1498, T1553.002 and T1021.002 to no D3FEND countermeasure (Demonstrated [85]). ATT&CK mitigation pages exist for M1051 Update Software, M1042 Disable or Remove Feature or Program, M1034 Limit Hardware Installation, M1037 Filter Network Traffic, M1049 Antivirus/Antimalware, M1017 User Training, M1050 Exploit Protection, M1021 Restrict Web-Based Content, M1041 Encrypt Sensitive Information, M1032 Multi-factor Authentication, M1027 Password Policies, M1036 Account Use Policies, M1048 Application Isolation and Sandboxing, M1038 Execution Prevention, M1045 Code Signing and M1026 Privileged Account Management (Demonstrated [86]). ICS-CERT's Stuxnet advice maps most closely to M1042 (disable AutoRun and shortcut icons), M1034 (USB caution), M1026 (least privilege), M1049 (current antivirus) and M1051 (the Siemens and Microsoft updates); MS10-046's gateway and SMB workarounds to M1037; and Gmail's default HTTPS to M1041 (Inferred [22][25][76]).

Navigator layers

The study's Enterprise techniques are saved as an ATT&CK Navigator layer pinned to ATT&CK v20, navigator/library-layers/chronicle-2010.json, in the format of the existing chronicle layers: domain enterprise-attack, described as a curated case-study layer and not an official MITRE mapping (Demonstrated [88]). It lists the 71 techniques in the Enterprise table under each tactic they carry in v20.0, 88 technique-tactic entries in all, and is listed on the Navigator layers page. The ten ICS techniques in the ATT&CK for ICS table are saved in a second layer, navigator/case-studies/chronicle-2010-ics.json, domain ics-attack, 12 technique-tactic entries, each scored 100 because ATT&CK lists the id for S0603; the home study's Stuxnet layer and its ICS layer carry all of Stuxnet. Scores show the basis, not severity, in the house scale set by the 2016 chronicle: 100 where an ATT&CK object prints the id, 70 where a primary source describes the behavior and this study assigns the id, 50 where only a Reported source supports it, and 40 for an Inferred mapping (Demonstrated [88]). The 58 techniques listed for G0066, S0203, C0002 or S0603 score 100, with a comment noting that Elderwood's tie to Aurora is "reportedly" and that Hydraq's techniques come from later analyses; T1102, T1539, T1110, T1566, T1056.001, T1657 and T1498 score 70; T1593.001 and T1566.003 score 50; T1505.003, T1550.004, T1040 and T1574.001 score 40. Each entry's comment names the source numbers in the Sources table below.

Cross-cutting themes

The headline events were disclosures. Aurora was detected in 2009 and disclosed in 2010; Buckshot Yankee happened in 2008 and was disclosed in 2010; Stuxnet's components date from 2008 and it was discovered in 2010 (Confirmed [1][38][34]).

Removable media crossed every boundary. A flash drive reached classified military networks in 2008 and a USB worm reached Siemens control software in 2010, both disclosed this year (Confirmed [38][22]).

Documents and browsers carried the attacks. Internet Explorer, Firefox, Reader and Flash flaws were exploited at or before their fixes from January to December (Confirmed [4][52][56][60][62][63][45]).

Encryption and integrity defaults moved. Gmail, HTTPS Everywhere, RFC 5746 and the signed DNS root all date from 2010, and Firesheep made the case in public (Confirmed [76][77][70][73][68]).

Courts did the takedowns. A civil restraining order cut off Waledac, police in Spain and Slovenia arrested Mariposa's suspected operators and creator, and Zeus cases ran in the US, UK, Ukraine and the Netherlands (Confirmed [15][16][43]).

The record named places, not perpetrators. Google said "originating from China", Shadows in the Cloud placed the attackers in Chengdu by circumstantial evidence, and the USCC assessed rather than found state involvement and could not tell what was done with the diverted traffic (Confirmed [1][18][10]).

Governments organized and exercised. Cyber Command began operations, Cyber Storm III ran, NIST moved to a life-cycle risk framework and smart grid guidance, and the UK funded a national programme (Confirmed [38][79][69][78][80]).

What is overstated and what is understated

This section is the study's reading of the record (Inferred).

- **Overstated: Aurora as one campaign with one exploit.** Google's post names no exploit, and ATT&CK ties Hydraq to Elderwood only through later reporting, with techniques from 2009 to 2012 [1][11][12].
- **Overstated: the number of companies hit by Aurora.** Google's post says "at least twenty other large companies"; the library's Notable Incidents line says "30+", a figure no source opened here gives, and names groups ATT&CK does not tie to Aurora [1][87].
- **Overstated: "Chinese state hacking" for Aurora, the Shadow network and the route event.** Google says "originating from China", the Shadows report places the attackers in Chengdu only circumstantially, and the USCC's state involvement is an assessment, not a finding [1][18][10].
- **Overstated: the route event as a hijack.** The USCC says the bad data appear to have originated at IDC China Telecommunication and that it cannot determine what, if anything, was done with the traffic; the 15 percent counts destinations, not traffic volume [10].
- **Overstated: Stuxnet as the "first cyberweapon" with novel lessons.** The 2010 defender advice was ordinary: removable-media policy, patching, least privilege and default credentials [22][28][33].
- **Overstated: the botnet and theft figures.** Mariposa's 8 to 12 million is an industry estimate, Waledac's 1.5 billion spam a day is Microsoft's estimate, and the USD 70 million is the FBI's aggregate across a ring, not a single theft [16][15][43].
- **Overstated: the precision of ids on 2010 events.** No ATT&CK mapping existed in 2010, every technique id here is assigned in hindsight, and KEV came 11 to 16 years later [5][8].
- **Understated: the KEV tail.** CVE-2010-0249 and CVE-2010-0806 entered KEV in 2026 and CVE-2010-3962 and CVE-2010-3765 in 2025 [8].
- **Understated: how the year's defaults spread.** The signed root, default HTTPS and RFC 5746 date from 2010; how fast they were adopted is outside the record opened here [73][76][70].
- **Understated: the AT&T exposure.** An unauthenticated web enumeration became a federal prosecution [20].

- **Understated: Buckshot Yankee.** The 2008 compromise was the most significant breach of US military computers, in the Deputy Secretary's words, and it appears in the 2010 record only as a disclosure [38].
- **Understated: the regulator record.** This study could not open the primaries for 2010 regulator and state data-security actions (listed under Scope and method), so it does not cover them.

Discrepancies in the record

- **When Stuxnet became public.** Symantec dates VirusBlokAda's report to 17 June; CERT/CC gives 10 July as the date public and 15 July as its first publication; Microsoft's advisory is dated 16 July [30][26][24].
- **ICSA-10-201's date.** The page shows only its 2014 revision, not an original release date [22].
- **Siemens' infection count.** Four infected customers in ICSA-10-201-01C; 15 infections worldwide on 7 September in ICSA-10-238-01B; about 100,000 infected hosts in Symantec's count, which is all hosts, not Siemens customers [22][28][30].
- **Stuxnet's vulnerabilities.** ICSA-10-238-01B says five, MS08-067 and four zero-days; ICSA-10-272-01 says four zero-days, two patched by then, plus MS08-067 [28][29].
- **CVE-2010-2568's weakness.** KEV lists CWE-20; NVD lists NVD-CWE-noinfo [8][27].
- **CVE-2010-3333's and CVE-2010-2883's weaknesses.** KEV lists CWE-119 for both; NVD lists CWE-787 for CVE-2010-3333 and the CISA ADP entry lists CWE-787 for CVE-2010-2883 [8][65][62].
- **CVE-2010-0188's publication date.** The CVE record gives 21 February; NVD gives 22 February [58][65].
- **Elderwood's techniques.** The live ATT&CK page shows T1027 as the parent row of T1027.002 and T1027.013; the library's data maps only the two sub-techniques to G0066 [11][84].
- **Butler's fraud figure.** The 12 February release counts more than 1.8 million card numbers; the 8 December release says more than 1.8 million cards with more than USD 87 million in fraud and more than 10,000 banks [13][14].
- **The Zeus totals.** SDNY's charges cover more than USD 3 million; the FBI's ring-wide figures are USD 70 million stolen and about USD 220 million attempted [42][43].
- **Shadows in the Cloud's date.** The report is dated 6 April; Citizen Lab's op-ed of 7 April calls it released "today" [18][19].
- **The KEV tail's range.** The first two CVE-2010 entries, SAP NetWeaver CVE-2010-5326 and JBoss Seam CVE-2010-1871, were added in November and December 2021, the month KEV began [8].
- **Aurora's attribution in the library.** Notable Incidents names "APT10 / Comment Crew (Chinese PLA Unit 61398, attributed)"; ATT&CK maps Aurora to Elderwood [87][11].

Sources

Every external source except source 82, which is carried from the E-Commerce Threats study, was opened on 12 November 2026, and the library data checks were run on 12 and 13 November 2026, as the Opened column shows. All are checked as of 13 November 2026. Grade is the source's evidence grade as used in this study. The address of source 82 is shown as code rather than as a link because sec.gov refuses automated link checkers; copy it into a browser to open it.

#	Source	URL	Opened	Grade
1	Google Official Blog, A new approach to China, 12 Jan 2010	https://googleblog.blogspot.com/2010/01/new-approach-to-china.html	2026-11-12	Confirmed
2	Google Official Blog, A new approach to China: an update, 22 Mar 2010	https://googleblog.blogspot.com/2010/03/new-approach-to-china-update.html	2026-11-12	Confirmed
3	Microsoft Security Advisory 979352, 14 Jan 2010	https://learn.microsoft.com/en-us/security-updates/securityadvisories/2010/979352	2026-11-12	Confirmed
4	Microsoft Security Bulletin MS10-002, 21 Jan 2010	https://learn.microsoft.com/en-us/security-updates/securitybulletins/2010/ms10-002	2026-11-12	Confirmed
5	NVD, CVE-2010-0249 (API record and detail page)	https://services.nvd.nist.gov/rest/json/cves/2.0?cveId=CVE-2010-0249 ; https://nvd.nist.gov/vuln/detail/CVE-2010-0249	2026-11-12	Confirmed
6	CVE Program, CVE-2010-0249 record	https://cveawg.mitre.org/api/cve/CVE-2010-0249	2026-11-12	Confirmed
7	CERT/CC, VU#492515, 14 Jan 2010	https://www.kb.cert.org/vuls/id/492515	2026-11-12	Confirmed
8	CISA, Known Exploited Vulnerabilities catalog (JSON feed, version 2026.11.10)	https://www.cisa.gov/sites/default/files/feeds/known_exploited_vulnerabilities.json	2026-11-12	Confirmed; Demonstrated (counts)
9	US Department of State (archived), Secretary Clinton, Remarks on Internet Freedom, 21 Jan 2010	https://2009-2017.state.gov/secretary/20092013clinton/rm/2010/01/135519.htm	2026-11-12	Confirmed
10	US-China Economic and Security Review Commission, 2010 Annual Report to Congress, Nov 2010	https://www.uscc.gov/sites/default/files/annual_reports/2010-Report-to-Congress.pdf	2026-11-12	Confirmed (the Commission's account and assessment); Reported (the vendor and Google statements it quotes)
11	MITRE ATT&CK, G0066 Elderwood	https://attack.mitre.org/groups/G0066/	2026-11-12	Confirmed (the object and its techniques); Reported (the Aurora attribution)
12	MITRE ATT&CK, S0203 Hydraq	https://attack.mitre.org/software/S0203/	2026-11-12	Confirmed
13	FBI Pittsburgh (USAO W.D. Pa.), Butler sentencing release, 12 Feb 2010	https://archives.fbi.gov/archives/pittsburgh/press-releases/2010/pt021210b.htm	2026-11-12	Confirmed
14	USAO W.D. Pa., release of 8 Dec 2010	https://www.justice.gov/archive/usao/paw/news/2010/2010_december/2010_12_08_01.html	2026-11-12	Confirmed

#	Source	URL	Opened	Grade
15	Microsoft, Cracking Down on Botnets, 24 Feb 2010 (On the Issues and Microsoft blog copies)	https://blogs.microsoft.com/on-the-issues/2010/02/24/cracking-down-on-botnets/ ; https://blogs.microsoft.com/blog/2010/02/24/cracking-down-on-botnets/	2026-11-12	Confirmed (the plaintiff's account of the order); Reported (Microsoft's estimates)
16	FBI, FBI, Slovenian and Spanish Police Arrest Mariposa Botnet Creator, Operators, 28 Jul 2010	https://archives.fbi.gov/archives/news/pressrel/press-releases/fbi-slovenian-and-spanish-police-arrest-mariposa-botnet-creator-operators	2026-11-12	Confirmed (arrests); Reported (the infection estimate)
17	Apache Infrastructure Team, apache.org incident report for 04/09/2010, 13 Apr 2010	https://infra.apache.org/blog/apache_org_04_09_2010	2026-11-12	Confirmed
18	Information Warfare Monitor and Shadowserver Foundation, Shadows in the Cloud: Investigating Cyber Espionage 2.0, JR03-2010, 6 Apr 2010 (mirror at nartv.org; Citizen Lab copy)	https://www.nartv.org/mirror/shadows-in-the-cloud.pdf ; https://citizenlab.ca/wp-content/uploads/2017/05/shadows-in-the-cloud.pdf	2026-11-12	Confirmed (findings); Reported (the Chengdu location, which the report calls circumstantial)
19	Citizen Lab, Breaking up dark clouds in cyberspace, 7 Apr 2010	https://citizenlab.ca/breaking-up-dark-clouds-in-cyberspace/	2026-11-12	Confirmed
20	FBI Newark (USAO D.N.J.), AT&T iPad charges, 18 Jan 2011	https://archives.fbi.gov/archives/newark/press-releases/2011/nk011811.htm	2026-11-12	Confirmed (as charged)
21	Microsoft Security Bulletin MS10-042, 13 Jul 2010	https://learn.microsoft.com/en-us/security-updates/securitybulletins/2010/ms10-042	2026-11-12	Confirmed
22	ICS-CERT, ICSA-10-201-01C, USB Malware Targeting Siemens Control Software (Update C)	https://www.cisa.gov/news-events/ics-advisories/icsa-10-201-01c	2026-11-12	Confirmed
23	US-CERT alert, Microsoft Windows .LNK Vulnerability, 16 Jul 2010 (CISA archive)	https://www.cisa.gov/news-events/alerts/2010/07/16/microsoft-windows-lnk-vulnerability	2026-11-12	Confirmed
24	Microsoft Security Advisory 2286198, 16 Jul 2010	https://learn.microsoft.com/en-us/security-updates/securityadvisories/2010/2286198	2026-11-12	Confirmed
25	Microsoft Security Bulletin MS10-046, 2 Aug 2010	https://learn.microsoft.com/en-us/security-updates/securitybulletins/2010/ms10-046	2026-11-12	Confirmed
26	CERT/CC, VU#940193, 15 Jul 2010	https://www.kb.cert.org/vuls/id/940193	2026-11-12	Confirmed
27	CVE Program and NVD, CVE-2010-2568 records	https://cveawg.mitre.org/api/cve/CVE-2010-2568 ; https://services.nvd.nist.gov/rest/json/cves/2.0?cveId=CVE-2010-2568	2026-11-12	Confirmed
28	ICS-CERT, ICSA-10-238-01B, Stuxnet Malware Mitigation (Update B)	https://www.cisa.gov/news-events/ics-advisories/icsa-10-238-01b	2026-11-12	Confirmed

#	Source	URL	Opened	Grade
29	ICS-CERT, ICSA-10-272-01, Primary Stuxnet Advisory, Sep 2010	https://www.cisa.gov/news-events/ics-advisories/icsa-10-272-01	2026-11-12	Confirmed
30	Symantec, W32.Stuxnet Dossier, first published 30 Sep 2010 (version 1.4, Feb 2011)	https://docs.broadcom.com/doc/security-response-w32-stuxnet-dossier-11-en	2026-11-12	Confirmed (vendor's analysis)
31	Microsoft Security Bulletin MS10-061, 14 Sep 2010	https://learn.microsoft.com/en-us/security-updates/securitybulletins/2010/ms10-061	2026-11-12	Confirmed
32	CVE Program, CVE-2010-2743 record (with the CVE-2010-2729 record as read)	https://cveawg.mitre.org/api/cve/CVE-2010-2743	2026-11-12	Confirmed
33	NVD, CVE-2010-2772 (API record)	https://services.nvd.nist.gov/rest/json/cves/2.0?cveId=CVE-2010-2772	2026-11-12	Confirmed
34	MITRE ATT&CK, S0603 Stuxnet	https://attack.mitre.org/software/S0603/	2026-11-12	Confirmed
35	World Nuclear News, Bushehr unaffected by computer virus, 27 Sep 2010 (relaying Reuters)	https://world-nuclear-news.org/Articles/Bushehr-unaffected-by-computer-virus	2026-11-12	Reported
36	Christian Science Monitor, Stuxnet: Ahmadinejad admits cyberweapon hit Iran nuclear program, 30 Nov 2010	https://www.csmonitor.com/USA/2010/1130/Stuxnet-Ahmadinejad-admits-cyberweapon-hit-Iran-nuclear-program	2026-11-12	Reported
37	Microsoft Security Advisory 2269637, 23 Aug 2010	https://learn.microsoft.com/en-us/security-updates/securityadvisories/2010/2269637	2026-11-12	Confirmed
38	William J. Lynn III, Defending a New Domain, Foreign Affairs, published 1 Sep 2010	https://www.foreignaffairs.com/articles/united-states/2010-09-01/defending-new-domain	2026-11-12	Confirmed (an official's own statement)
39	Microsoft Security Advisory 2416728, 17 Sep 2010	https://learn.microsoft.com/en-us/security-updates/securityadvisories/2010/2416728	2026-11-12	Confirmed
40	Microsoft Security Bulletin MS10-070, 28 Sep 2010	https://learn.microsoft.com/en-us/security-updates/securitybulletins/2010/ms10-070	2026-11-12	Confirmed
41	Twitter (X) blog, All about the "onMouseOver" incident, 21 Sep 2010	https://blog.x.com/en/company/all-about-the-onmouseover-incident	2026-11-12	Confirmed
42	FBI New York (USAO S.D.N.Y.), 37 defendants charged, 30 Sep 2010	https://archives.fbi.gov/archives/newyork/press-releases/2010/nyfo093010.htm	2026-11-12	Confirmed (as charged)
43	FBI story, Cyber Banking Fraud, 1 Oct 2010	https://archives.fbi.gov/archives/news/stories/2010/october/cyber-banking-fraud	2026-11-12	Confirmed

#	Source	URL	Opened	Grade
44	FBI, International Cooperation Disrupts Multi-Country Cyber Theft Ring, 1 Oct 2010	https://archives.fbi.gov/archives/news/pressrel/press-releases/international-cooperation-disrupts-multi-country-cyber-theft-ring	2026-11-12	Confirmed
45	Microsoft Security Advisory 2458511, 3 Nov 2010 (resolved by MS10-090, 14 Dec 2010)	https://learn.microsoft.com/en-us/security-updates/securityadvisories/2010/2458511	2026-11-12	Confirmed
46	FBI and DOJ, Sixteen Individuals Arrested in the United States for Alleged Roles in Cyber Attacks, 19 Jul 2011	https://archives.fbi.gov/archives/news/pressrel/press-releases/sixteen-individuals-arrested-in-the-united-states-for-alleged-roles-in-cyber-attacks	2026-11-12	Confirmed (as charged)
47	MITRE ATT&CK, C0002 Night Dragon	https://attack.mitre.org/campaigns/C0002/	2026-11-12	Confirmed (the record); Reported (the attribution)
48	MITRE ATT&CK, S0143 Flame	https://attack.mitre.org/software/S0143/	2026-11-12	Confirmed
49	Microsoft Security Bulletin MS10-106, 14 Dec 2010 (the 2010 series, MS10-001 to MS10-106)	https://learn.microsoft.com/en-us/security-updates/securitybulletins/2010/ms10-106	2026-11-12	Confirmed (dates); Inferred (count)
50	Microsoft Security Bulletin MS10-071, 12 Oct 2010 (the October bulletins, MS10-071 to MS10-086)	https://learn.microsoft.com/en-us/security-updates/securitybulletins/2010/ms10-071	2026-11-12	Confirmed (dates); Inferred (counts)
51	Microsoft Security Bulletin MS10-015, 9 Feb 2010	https://learn.microsoft.com/en-us/security-updates/securitybulletins/2010/ms10-015	2026-11-12	Confirmed
52	Microsoft Security Bulletin MS10-018, 30 Mar 2010	https://learn.microsoft.com/en-us/security-updates/securitybulletins/2010/ms10-018	2026-11-12	Confirmed
53	Microsoft Security Bulletin MS10-073, 12 Oct 2010	https://learn.microsoft.com/en-us/security-updates/securitybulletins/2010/ms10-073	2026-11-12	Confirmed
54	Microsoft Security Bulletin MS10-087, 9 Nov 2010	https://learn.microsoft.com/en-us/security-updates/securitybulletins/2010/ms10-087	2026-11-12	Confirmed
55	Microsoft Security Bulletin MS10-092, 14 Dec 2010	https://learn.microsoft.com/en-us/security-updates/securitybulletins/2010/ms10-092	2026-11-12	Confirmed
56	Mozilla Foundation Security Advisory 2010-73, 27 Oct 2010	https://www.mozilla.org/en-US/security/advisories/mfsa2010-73/	2026-11-12	Confirmed
57	CVE Program, CVE-2010-3765 record	https://cveawg.mitre.org/api/cve/CVE-2010-3765	2026-11-12	Confirmed
58	CVE Program, CVE-2010-0188 record	https://cveawg.mitre.org/api/cve/CVE-2010-0188	2026-11-12	Confirmed
59	CVE Program, CVE-2010-0738 record	https://cveawg.mitre.org/api/cve/CVE-2010-0738	2026-11-12	Confirmed

#	Source	URL	Opened	Grade
60	CVE Program, CVE-2010-1297 record	https://cveawg.mitre.org/api/cve/CVE-2010-1297	2026-11-12	Confirmed
61	CVE Program, CVE-2010-2861 record	https://cveawg.mitre.org/api/cve/CVE-2010-2861	2026-11-12	Confirmed
62	CVE Program, CVE-2010-2883 record	https://cveawg.mitre.org/api/cve/CVE-2010-2883	2026-11-12	Confirmed
63	CVE Program, CVE-2010-3654 record	https://cveawg.mitre.org/api/cve/CVE-2010-3654	2026-11-12	Confirmed
64	CVE Program, CVE-2010-4344 record (with the CVE-2010-4398 and CVE-2010-3904 records as read)	https://cveawg.mitre.org/api/cve/CVE-2010-4344	2026-11-12	Confirmed
65	NVD, CVE-2010-3333 (API record; with CVE-2010-0188 as read)	https://services.nvd.nist.gov/rest/json/cves/2.0?cveId=CVE-2010-3333	2026-11-12	Confirmed
66	USAO D. Mass., Toey sentencing release, 16 Apr 2010 (records the Gonzalez sentence)	https://www.justice.gov/archive/usao/ma/news/2010/April/TOEYsentPR.html	2026-11-12	Confirmed
67	USAO D. Mass., Scott sentencing release, 29 Mar 2010	https://www.justice.gov/archive/usao/ma/news/2010/March/ScottSentencingPR.html	2026-11-12	Confirmed
68	Eric Butler, Firesheep, 24 Oct 2010	https://codebutler.com/2010/10/24/firesheep/	2026-11-12	Confirmed (the author's post); Demonstrated (the tool)
69	NIST, SP 800-37 Revision 1 (original and update 1)	https://csrc.nist.gov/pubs/sp/800/37/r1/final ; https://csrc.nist.gov/pubs/sp/800/37/r1/upd1/final	2026-11-12	Confirmed
70	IETF, RFC 5746, Transport Layer Security (TLS) Renegotiation Indication Extension, Feb 2010	https://www.rfc-editor.org/rfc/rfc5746	2026-11-12	Confirmed
71	MITRE and SANS, 2010 CWE/SANS Top 25 Most Dangerous Software Errors	https://cwe.mitre.org/top25/archive/2010/2010_cwe_sans_top25.html	2026-11-12	Confirmed
72	OWASP, OWASP Top 10 2010 (PDF)	https://owasp.org/www-pdf-archive/OWASP_Top_10_-_2010.pdf	2026-11-12	Confirmed
73	IANA, root zone trust anchors (root-anchors.xml)	https://data.iana.org/root-anchors/root-anchors.xml	2026-11-12	Confirmed
74	Chromium Blog, Encouraging more Chromium security research, 28 Jan 2010	https://blog.chromium.org/2010/01/encouraging-more-chromium-security.html	2026-11-12	Confirmed
75	Mozilla Security Blog, Refresh, 15 Jul 2010	https://blog.mozilla.org/security/2010/07/15/refresh/	2026-11-12	Confirmed
76	Official Gmail Blog, Default https access for Gmail, 12 Jan 2010	https://gmail.googleblog.com/2010/01/default-https-access-for-gmail.html	2026-11-12	Confirmed

#	Source	URL	Opened	Grade
77	EFF, Encrypt the Web with the HTTPS Everywhere Firefox Extension, 17 Jun 2010	https://www.eff.org/deeplinks/2010/06/encrypt-web-https-everywhere-firefox-extension	2026-11-12	Confirmed
78	NIST, NISTIR 7628, Guidelines for Smart Grid Cyber Security, 31 Aug 2010	https://csrc.nist.gov/pubs/ir/7628/final	2026-11-12	Confirmed
79	DHS, Cyber Storm III Final Report	https://www.cisa.gov/sites/default/files/publications/CyberStorm%2520III%2520FINAL%2520Report.pdf	2026-11-12	Confirmed
80	UK Government, The Strategic Defence and Security Review: Securing Britain in an Age of Uncertainty, 19 Oct 2010	https://www.gov.uk/government/publications/the-strategic-defence-and-security-review-securing-britain-in-an-age-of-uncertainty	2026-11-12	Confirmed
81	The White House (archived), CNCI summary page (national-initiative)	https://obamawhitehouse.archives.gov/issues/foreign-policy/cybersecurity/national-initiative	2026-11-12	Confirmed (content); date not checked
82	Heartland Payment Systems, Exhibit 99.1 on the Visa settlement, 8 Jan 2010 (SEC EDGAR; as cited by the E-Commerce Threats study, not reopened)	`https://www.sec.gov/Archives/edgar/data/0001144354/000119312510003149/dex991.htm`	not opened (carried from E-Commerce Threats source 6, opened there 2026-11-11)	Inferred (carried from the home study)
83	TeamStarWolf Intelligence Hub, feed (used as a map)	https://teamstarwolf.github.io/intelligence-hub/data/feed.json	2026-11-12	Confirmed (as feed data)
84	TeamStarWolf library, ATT&CK data files (`data/attack/`, `data/attack/ics/`)	https://github.com/TeamStarWolf/TeamStarWolf/tree/main/data/attack	2026-11-12; 2026-11-13	Confirmed; Demonstrated
85	TeamStarWolf library, weakness and countermeasure data (`data/weaknesses/capec.jsonl`, `cwe.jsonl`, `data/attack/technique_to_d3fend.jsonl`)	https://github.com/TeamStarWolf/TeamStarWolf/tree/main/data/weaknesses	2026-11-12; 2026-11-13	Demonstrated
86	TeamStarWolf library, `mitre/` page tree (technique, group, software, campaign, CAPEC, D3FEND and mitigation pages)	https://github.com/TeamStarWolf/TeamStarWolf/tree/main/mitre	2026-11-12; 2026-11-13	Demonstrated
87	TeamStarWolf library, case studies on main (home studies named in the text; Notable Incidents line 84 as flagged by the Hacker and the State study)	https://github.com/TeamStarWolf/TeamStarWolf/tree/main/case-studies	2026-11-12; 2026-11-13	Confirmed (for the fact that a study is the home and for the facts it carries from its own sources)
88	TeamStarWolf library, chronicle Navigator layers (`navigator/library-layers/chronicle-2016.json` and the later years)	https://github.com/TeamStarWolf/TeamStarWolf/tree/main/navigator/library-layers	2026-11-13	Demonstrated

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The terms for every MITRE dataset used here are recorded in THIRD_PARTY_NOTICES.md.

Related references

- Notable Incidents, Threat Actors, Malware Families, CVE Reference
- CWE Reference, CAPEC Reference, D3FEND Reference, Cryptography, Browser Security, Email Security, GRC and Compliance, Vulnerability Management, Incident Response, ICS/OT Security
- Home studies: The Zero-Day Trade and Its Fallout; Stuxnet; Four Books on Cybercrime; E-Commerce Threats; China-attributed threat actors; United States-attributed threat actors; The Hacker and the State; APT41; ICS attacks and advisories; Log4Shell; SolarWinds SUNBURST; npm supply-chain attacks; Hugging Face supply chain
- Series: 1993, 1994, 1995, 1996, 1997, 1998, 1999, 2000, 2001, 2002, 2003, 2004, 2005, 2006, 2007, 2016, 2017, 2018, 2019, 2020, 2021, 2022, 2023, 2024, 2025, 2026
- Techniques: T1189 Drive-by Compromise, T1203 Exploitation for Client Execution, T1566.001 Spearphishing Attachment, T1566.002 Spearphishing Link, T1566.003 Spearphishing via Service, T1593.001 Social Media, T1091 Replication Through Removable Media, T1210 Exploitation of Remote Services, T1068 Exploitation for Privilege Escalation, T1014 Rootkit, T1553.002 Code Signing, T1078.001 Default Accounts, T1190 Exploit Public-Facing Application, T1102 Web Service, T1539 Steal Web Session Cookie, T1110 Brute Force, T1505.003 Web Shell, T1040 Network Sniffing, T1566 Phishing, T1056.001 Keylogging, T1657 Financial Theft, T1498 Network Denial of Service, T1574.001 DLL; the full list is in the Part 9 table
- CAPEC: CAPEC-21, CAPEC-31, CAPEC-55, CAPEC-62, CAPEC-63, CAPEC-65, CAPEC-66, CAPEC-68, CAPEC-70, CAPEC-94, CAPEC-98, CAPEC-100, CAPEC-102, CAPEC-112, CAPEC-126, CAPEC-163, CAPEC-191, CAPEC-457, CAPEC-471, CAPEC-552, CAPEC-555, CAPEC-560, CAPEC-561, CAPEC-562, CAPEC-568, CAPEC-641, CAPEC-644, CAPEC-650
- Groups, software and campaigns: Elderwood G0066; Axiom G0001; Hydraq S0203; PoisonIvy S0012; Stuxnet S0603; Flame S0143; ASPXSpy S0073; at S0110; gsecdump S0008; PsExec S0029; zwShell S0350; Night Dragon C0002; Operation Dust Storm C0016
- Navigator layers, including this study's layers, chronicle-2010.json and chronicle-2010-ics.json